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# FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA

PHY 113 — Past Questions & Materials

**by Renance**

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Course: PHY 113

Institution: Federal University of Technology, Minna

Papers in this pack: 1

Levels: 100

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# Paper 1 of 1 — PHY113 handout

Level 100 • First Semester • Handout

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## PHY 113: Study Guide

### Introduction

PHY 113: Mechanics is a 2-credit unit course for students studying towards acquiring a Bachelor of Technology in Information Technology and other related disciplines. The course is divided into 5 modules and 17 study units.

### Study Units

There are 17 study units and 5 modules in this course. They are:

#### Units and Measurements

Module One Unit 1 Units and dimensions

Unit 2 Space and time

#### Motion and Forces

Unit 1 Scalars and Vectors

Module Two Unit 2 Scalars and Vectors

Unit 3 Newton's laws of motion

Unit 4 Collision, momentum and Impulse

Unit 5 Work, Energy and Power

#### Circular Motion

Module Three Unit 1 Centripetal Force

Unit 2 Newtonian Gravitation

Unit 3 Circular Motion in Space

#### Rotation of Rigid bodies

Unit 1 Rotational Variables

#### Module Four

Unit 2 Moment of Inertia

Unit 3 Conservation of Angular Momentum

#### Statics

Unit 1 Composition and resolution of forces

#### Module Five

Unit 2 Concurrent and parallel forces

Unit 3 Conditions for Equilibrium

Centre of Gravity for Component Bodies and bodies

Unit 4

from which part is removed

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#### Module 1

Units and Measurement

Unit 1. Units and Dimensions

Unit 2. Space and Time

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## Unit 1

## Units and Dimensions

## Content

## 1.0 Introduction

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## 3.0 Main Content

## 3.1 Scope of Physics

## 3.2 Units and Measurements

## 3.3 The International System of Units

## 3.4 Units Consistency and Conversion

## 3.5 Dimensional Analysis of the Equation

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## 1.0 Introduction

This unit introduces to you a very important aspect of physics, which is also one of the most fundamentals of sciences. Scientists of all disciplines use the ideas of physics, including chemists who study the structure of molecules, palaeontologists who try to reconstruct how dinosaurs walked, and climatologists who study how human activities affect the atmosphere and oceans. It is also the foundation of all engineering and technology.

## 2.0 Objectives

At the end of this unit, you should be able to:

- i. Describe the scope of physics.
- ii. Explain the concept of fundamental and derived quantities and their units.
- iii. Learn three fundamental quantities of mechanics and their units.

## 3.0 Main Content

## 3.1 Scope of Physics

Do you know that the knowledge of physics is useful in everyday situations as well as in non-scientific professions? It can help you understand how microwave ovens work, why metals should not be put into them, and why they might affect pacemakers. It also allows you to understand the hazards of radiation and to evaluate these hazards rationally and more easily.

Physics also explains the reason why a black car radiator helps remove heat in a car engine, and it explains why a white roof helps keep the inside of a house cool. Similarly,

the operation of a car's ignition system as well as the transmission of electrical signals throughout our body's nervous system are much easier to understand when you think about them in terms of basic physics

Physics is involved in medical diagnostics, such as radiographs, magnetic resonance imaging, and ultrasonic blood flow measurements. Medical therapy sometimes involves physics directly; for example, cancer radiotherapy uses ionizing radiation. Physics also explains sensory phenomena, such as how musical instruments make sound, how the eye detects color, and how lasers transmit information. It is not necessary to study all applications of physics formally. What is most useful is knowing the basic laws of physics and developing skills in the analytical methods for applying them. The study of physics also can improve your problem solving skills. Furthermore, physics retains the most basic

aspects of science, so it is used by all the sciences, and the study of physics makes other sciences easier to understand.

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#### Self-Assessment Exercise 1

Does physics explains the reason why a black car radiator helps remove heat in a car engine and why a white roof helps keep the inside of a house cool.

A. Yes B. False

#### 3.2 Units and Measurements

The comparison of any physical quantity with its standard unit is called measurement.

##### Physical Quantities

Do you know that all quantities in terms of which laws of physics are described, and whose measurement is necessary are called physical quantities?

##### Units

- A definite amount of a physical quantity is taken as its standard unit.
- The standard unit should be easily reproducible and internationally accepted.

##### Fundamental Units

Those physical quantities that are independent to each other are called fundamental quantities and their units are called fundamental units.

The three fundamental quantities in mechanics and their respective SI units are:

Mass, S.I Unit = Kilogram (kg)

Length, S.I Unit = Meter (m)

Time, S.I Unit = Seconds (s)

##### Derived Units

Those physical quantities which are obtained from a combination of two or the three fundamental quantities are called derived quantities and their units are called derived units. e.g., velocity, acceleration, force, work etc.

#### Self-Assessment Exercise(s) 2

1. Which of the following are fundamental units?

a. Kg b. Kg.m-3 c. s d. m<sup>3</sup> e. Nm<sup>-2</sup>

2. Which of the units of the following physical quantities are derived?

a. Area b. thrust c. Pressure d. Mass

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### 3.3 The International System of Units

In 1960, an international committee established a set of standards for length, mass, and other basic quantities. The system established is an adaptation of the metric system, and it is called the SI system of units. (The abbreviation SI comes from the system's French name "Système International.") In this system, the units of length, mass, and time are the meter, kilogram, and second, respectively. Other SI standards established by the committee are those for temperature (the kelvin), electric current (the ampere), luminous intensity (the candela), and the amount of substance (the mole). These are generally referred to as the fundamental quantities and units.

Many SI derived units are defined in terms of these base units. For example, the SI unit for power, called the watt (W), is defined in terms of the base units for mass, length and time.

$$1 \text{ kg m}^2 \text{ s}^{-3} = 1 \text{ W} = 1 \text{ kg} \cdot \text{m}^2 / \text{s}^3,$$

Where the last collection of unit symbols is read as kilogram-meter squared per second cubed. To express the very large and very small quantities we often run into in physics, we use scientific notation, which employs powers of 10. In this notation,

$$3560\,000\,000 = 3.56 \times 10^9$$

$$\text{and } 0.000\,000\,4925 = 4.92 \times 10^{-7}$$

As a further convenience when dealing with very large or very small measurements, we use the prefixes listed in Table 1.1.

Table 1.1 Metric Prefixes for Powers of 10 and Their Symbols

Factor Prefix Symbol

10<sup>24</sup> yotta- Y

10<sup>21</sup> zetta- Z

10<sup>18</sup> exa- E

10<sup>15</sup> peta- P

10<sup>12</sup> tera- T

10<sup>9</sup> giga- G

10<sup>6</sup> mega- M

10<sup>3</sup> kilo- K

10<sup>2</sup> hecto- h

10<sup>1</sup> deka- da

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10<sup>-1</sup> deci- d

10<sup>-2</sup> centi- c

10<sup>-3</sup> milli- m

10–6 micro-  $\mu$   
 10–9 nano- n  
 10–12 pico- p  
 10–15 femto- f  
 10–18 atto- a  
 10–21 zepto- z  
 10–24 yocto- y

Attaching a prefix to an SI unit has the effect of multiplying by the associated factor. Thus, we can express a particular electric power as

$$1.28 \times 10^9 \text{ W} = 1.28 \text{ GW} = 1.28 \text{ kW}$$

We shall now discuss briefly the three fundamental quantities of mechanics.

### Length

In A.D. 1120 the king of England decreed that the standard of length in his country would be named the yard and would be precisely equal to the distance from the tip of his nose to the end of his outstretched arm. Similarly, the original standard for the foot adopted by the French was the length of the royal foot of King Louis XIV. This standard prevailed until 1799, when the legal standard of length in France became the meter, defined as one ten-millionth the distance from the equator to the North Pole along one particular longitudinal line that passes through Paris. Many other systems for measuring length have been developed over the years, but the advantages of the French system have caused it to prevail in almost all countries and in scientific circles everywhere. As recently as 1960, the length of the meter was defined as the distance between two lines on a specific platinum-iridium bar stored under controlled conditions in France. This standard was abandoned for several reasons, a principal one being that the limited accuracy with which the separation between the lines on the bar can be determined does not meet the current requirements of science and technology. In the 1960s and 1970s, the meter was defined as 1 650 763.73 wavelengths of orange-red light emitted from a krypton-86 lamp. However, in October 1983, the meter (m) was redefined as the distance traveled by light in vacuum during a time of  $1/299\,792\,458$  second. In effect, this latest definition establishes that the speed of light in vacuum is precisely 299 792 458 m per second. Other recognized units of length are the nanometers (nm), millimeter (mm), Centimeter (cm) and the Kilometer (km). You can convert from one unit to another, using the factors

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provided in Table 1.1. It is however required that physical quantities are expressed in the SI unit system.

Table 1.2 Approximate values of some measured lengths.

Approximate Values of Some Measured Lengths Length(m)

Distance from the Earth to most remote known quasar  $1.4 \times 10^{26}$

Distance from the Earth to most remote known normal galaxies  $9 \times 10^{25}$

Distance from the Earth to nearest large galaxy (M 31, the Andromeda galaxy)  $2 \times 10^{22}$

Distance from the Sun to nearest star (Proxima Centauri)  $4 \times 10^{16}$

One Lightyear  $9.46 \times 10^{15}$

Mean orbit radius of the Earth about the Sun  $1.50 \times 10^{11}$

Mean Distance from the Earth to the Moon  $3.84 \times 10^8$

Distance from the equator to the North Pole  $1.100 \times 10^7$

Mean Radius of the Earth  $6.37 \times 10^6$

Typical Altitude (above the surface) of a satellite orbiting the Earth  $2 \times 10^5$

Length of a Football Field  $9.1 \times 10^1$

Length of a Housefly  $5 \times 10^{-3}$

Size of Small Dust Particles  $\sim 10^{-4}$

Size of Cells of most living organisms  $\sim 10^{-5}$

Diameter of a Hydrogen Atom  $\sim 10^{-10}$

Diameter of an Atomic Nucleus  $\sim 10^{-14}$

Diameter of a Proton  $\sim 10^{-15}$

#### Mass

The standard of mass, the kilogram (abbreviated kg), is defined to be the mass of a particular cylinder of platinum-iridium alloy kept at the International Bureau of Weights and Measures at Sèvres, near Paris. An atomic standard of mass would be more fundamental, but at present we cannot measure masses on an atomic scale with as much accuracy as on a macroscopic scale. The gram (which is not a fundamental unit) is 0.001 kilogram. Other recognized units of mass include the microgram ( $\mu\text{g} = 10^{-9}\text{kg}$ ), the milligram ( $\text{mg} = 10^{-6}\text{kg}$ )

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#### Table 1.3 Approximate values of the masses of various objects

##### Masses of Various Bodies (Approx. Values)

##### Body Mass (kg)

Visible Universe  $\sim 10^{52}$

Milky Way Galaxy  $7 \times 10^{41}$

Sun  $1.99 \times 10^{30}$

Earth  $5.98 \times 10^{24}$

Moon  $7.36 \times 10^{22}$

Horse  $\sim 10^3$

Human  $\sim 10^2$

Frog  $\sim 10^{-1}$

Mosquito  $\sim 10^{-5}$

Bacterium  $\sim 10^{-15}$

Hydrogen Atom  $1.67 \times 10^{-27}$

Electron  $9.11 \times 10^{-31}$

Time

The SI unit for time, the second (abbreviated s), has a long history. For many years, it was defined as  $1/86,400$  of a mean solar day. More recently, a new standard was adopted to gain greater accuracy and to define the second in terms of a non varying or constant physical phenomenon (because the solar day is getting longer because of the very gradual slowing of Earth's rotation). Cesium atoms can be made to vibrate in a very steady way, and these vibrations can be readily observed and counted. In 1967, the second was redefined as the time required for 9,192,631,770 of these vibrations to occur. Other units of time are minutes (min = 60sec), hour (hr = 3600sec), day (dy = 86,400sec), year (yr =  $3.15 \times 10^7$ sec)

### Example 1.1

Using metric prefixes;

Restate the mass  $1.93 \times 10^{13}$  kg using a metric prefix such that the resulting numerical value is bigger than one but less than 1000.

Solution

Replacing the k in kilogram with the factor of  $10^3$ , we find that

$$1.93 \times 10^{13} \text{ kg} = 1.93 \times 10^{13} \times 10^3 \text{ g} = 1.93 \times 10^{16} \text{ g}.$$

From table 1.1, we see that  $10^{16}$  is between "peta-"  $10^{15}$  and "exa-"  $10^{18}$ . If we use the "peta-" prefix, then we find that  $1.93 \times 10^{16} \text{ g} = 1.93 \times 10^1 \text{ Pg}$ , since  $16 = 1 + 15$ .

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Alternatively, if we use the "exa-" prefix we find that  $1.93 \times 10^{16} = 1.93 \times 10^{-2} \text{ Eg}$ , since  $16 = -2 + 18$ . Because the problem asks for the numerical value between one and 1000, we use the "peta-" prefix and the answer is 19.3 Pg.

Self-Assessment Exercise(s) 3

1. Which of the following is NOT a fundamental S.I unit?

a. Meter b. Ampere c. Kelvin d. Second e. Radian

2. The internationally agreed system of units (SI) for physical measurements are

a. lb,ft,sec b. g,m,sec c. kg, m, sec d. kg, cm, sec

### 3.4 Unit Consistency and Conversion

We use equations to express relationships among physical quantities, represented by algebraic symbols. Each algebraic symbol always denotes both a number and a unit. For example, d might represent a distance of 10m, t a time of 5s, and a speed of  $2 \text{ m/s}$

An equation must always be dimensionally consistent. You can't add apples and automobiles; two terms may be added or equated only if they have the same units. For example, if a body moving with constant speed travels a distance d in a time t, these quantities are related by the equation  $d = vt$

If d is measured in meters, then the product must also be expressed in meters. Using the

above numbers as an example, we may write  $10 \text{ m} = (2 \text{ m/s}) (5 \text{ s})$

□

Because the unit on the right side of the equation cancels the unit s, the product has units of meters, as it must. In calculations, units are treated just like algebraic symbols with respect to multiplication and division.

### Example 1.2

Ade runs 100□ in 9.58□. What is his speed in m/s?

Solution:

□ 100□

Speed = □ = 9.58□ = 10.4 □□

### 3.5 Dimensional Analysis of Equation

The word dimension has a special meaning in physics. It usually denotes the physical nature of a quantity. Whether a distance is measured in the length unit feet or the length unit meters, it is still a distance. We say the dimension—the physical nature—of distance is length. The symbols we use in this book to specify the three fundamental quantities in mechanics, length, mass, and time are L, M, and T, respectively. We shall often use brackets [ ] to denote the dimensions of a physical quantity. Thus, the dimension of the

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three fundamental quantities are [L], [M], and [T]. For example, the symbol we use for speed is v, and in our notation the dimensions of speed are written As another example, the dimensions of area, for which we use the symbol A, are The dimensions of area, volume, speed, and acceleration are listed in Table 1.4. In solving problems in physics, there is a useful and powerful procedure called dimensional analysis. This procedure, which should always be used, will help minimize the need for rote memorization of equations. Dimensional analysis makes use of the fact that dimensions can be treated as algebraic quantities. That is, quantities can be added or subtracted only if they have the same dimensions. Furthermore, the terms on both sides of an equation must have the same dimensions.

Table 1.4 Dimension and common units of Area, Volume, Speed and Acceleration

System Area [L]<sup>2</sup> Volume [L]<sup>3</sup> Speed [LT<sup>-1</sup>] Acceleration [LT<sup>-2</sup>]

SI □<sup>2</sup> □<sup>3</sup> □/□ □/□<sup>2</sup>

British □□<sup>2</sup> □□<sup>3</sup> □□/□ □□/□<sup>2</sup>

engineering

Area = Length x Breath

= [L] x [L] = [L]<sup>2</sup>

Volume = Length x Breath x Height

= [L] x [L] x [L] = [L]<sup>3</sup>

□□□□□□□□ [□]

Speed = = = [□□<sup>-1</sup>]

□□□□ [□]

Acceleration =

Acceleration =

Acceleration =

Acceleration =

Acceleration =

Acceleration =

By following these simple rules, you can use dimensional analysis to help determine whether an expression has the correct form. The relationship can be correct only if the dimensions are the same on both sides of the equation. To illustrate this procedure, suppose you wish to derive a formula for the distance  $x$  traveled by a train in a time  $t$  if the train starts from rest and moves with constant acceleration  $a$ . We shall find that the

correct expression is  $x = \frac{1}{2} a t^2$ . Let us use dimensional analysis to check the validity of

this expression. The quantity  $x$  on the left side has the dimension of length. For the equation to be dimensionally correct, the quantity on the right side must also have the dimension of length. We can perform a dimensional check by substituting the dimensions

for acceleration,  $[L T^{-2}]$ , and time,  $[T]$ , into the equation. That is, the dimensional form of

the equation  $x = \frac{1}{2} a t^2$  is

$[L]$

$[L] = \frac{1}{2} [L T^{-2}] [T]^2 = [L]$

$[L]$

The units of time squared cancel as shown, leaving the unit of length. A more general procedure using dimensional analysis is to set up an expression of the form  $x \propto a^n t^m$  where  $n$  and  $m$  are exponents that must be determined and the symbol  $\propto$  indicates a proportionality. This relationship is correct only if the dimensions of both sides are the same. Because the dimension of the left side is length, the dimension of the right side must also be length. That is,  $[L] = [L^n T^{-2n} T^m] = [L^n T^{m-2n}]$

Because the dimensions of acceleration are  $L/T^2$  and the dimension of time is  $T$ , we

have

$[L] = [L^n T^{-2n} T^m] = [L^n T^{m-2n}]$  the exponents of  $L$  and  $T$  must be the same on both

sides, the dimensional equation is balanced under the conditions  $n - 2n = 0$ ,  $n = 1$  and  $m = 2$ . Returning to our original expression  $x \propto a^n t^m$ , we conclude that  $x \propto a t^2$ . This

result differs by a factor of 2 from the correct expression, which is  $x = \frac{1}{2} a t^2$ . Because the

factor is dimensionless, there is no way of determining it using dimensional analysis.

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## Self-Assessment Exercise 4

1. In dimensional analysis quantities can be added or subtracted only if they have the same dimensions?

True or False

## 4.0 Conclusion

This unit took you through the basic concept of unit and measurement, brief history of SI unit and the overview of Dimensional analysis

## 5.0 Summary

The three fundamental physical quantities of mechanics are length, mass, and time, which in the SI system have the units meters (m), kilograms (kg), and seconds (s), respectively. Prefixes indicating various powers of ten are used with these three basic units. Derived units for other physical quantities are products or quotients of the basic units. Equations must be dimensionally consistent that is two terms can be added only when they have the same units.

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## 6.0 Tutor Marked Assignment

1. Describe the scope of physics.
2. Explain the concept of basic and derived quantities.
3. Mention the fundamental quantities of physics and the units physicists use to measure them.

## 7.0 References/Further Reading

Fundamentals of physics (10th Edition) Jearl Walker, 2014 University physics with modern physics(13th Edition) Hugh D. Young and Roger A. Freedman,2012.  
University physics, Volume 1 . Samuel J. Ling, Jeff Sanny,and William Moebs, 2018.

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## Unit 2

## Time and Space

## Content

## 1.0 Introduction

## 2.0 Objectives

## 3.0 Main Content

## 3.1 General Overview on Space and Time

## 3.2 Newton's Views on Space and Time

## 3.3 Frames of Reference

## 3.4 Inertial Frames of Refernce

## 4.0 Conclusion

## 5.0 Summary

## 6.0 Tutor Marked Assignment

## 7.0 References/Further Reading

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### 1.0 Introduction

This unit introduces you to Einstein special theory of relativity, proposing drastic revisions in the Newtonian concepts of space and time. The special theory of relativity has made wide-ranging changes in our understanding of nature, but Einstein based it on just two simple postulates. One states that the laws of physics are the same in all inertial frames of reference; the other states that the speed of light in vacuum is the same in all inertial frames.

### 2.0 Objectives

At the end of this unit, you should be able to:

1. Explain in clear term the principle of space and time.
2. Define frame of reference
3. Describe Einstein's postulates of frame of reference.

### 3.0 Main Content

#### 3.1 General Overview on Space and Time

Space is the boundless three-dimensional extent in which objects and events have relative position and direction. Physical space is often conceived in three linear dimensions, although modern physicists usually consider it, with time, to be part of a boundless four-dimensional continuum known as spacetime. The concept of space is considered to be of fundamental importance to an understanding of the physical universe. However, disagreement continues between philosophers over whether it is itself an entity, a relationship between entities, or part of a conceptual framework. In Isaac Newton's view, space was absolute in the sense that it existed permanently and independently, whether there was any matter in the space. Other natural philosophers, notably Gottfried Leibniz, thought instead that space was in fact a collection of relations between objects, given by their distance and direction from one another. In the 18th century, the philosopher and theologian George Berkeley attempted to refute the "visibility of spatial depth" in his Essay Towards a New Theory of Vision. Later, the metaphysician Immanuel Kant said that the concepts of space and time are not empirical ones derived from experiences of the outside world, they are elements of an already given systematic framework that humans possess and use to structure all experiences.

In the 19th and 20th centuries mathematicians began to examine geometries that are non-Euclidean, in which space is conceived as curved, rather than flat. According to Albert Einstein's theory of general relativity, space around gravitational fields deviates from

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Euclidean space. Experimental tests of general relativity have confirmed that non-Euclidean geometries provide a better model for the shape of space.

Time is the indefinite continued progress of existence and events that occur in an apparently irreversible succession from the past, through the present, to the future. Time is a component quantity of various measurements used to sequence events, to compare the duration

of events or the intervals between them, and to quantify rates of change of quantities in material reality or in the conscious experience. Time is often referred to as a fourth dimension, along with three spatial dimensions.

Time has long been an important subject of study in religion, philosophy, and science, but defining it in a manner applicable to all fields without circularity has consistently eluded scholars.

Nevertheless, diverse fields such as business, industry, sports, the sciences, and the performing arts all incorporate some notion of time into their respective measuring systems.

Time in Physics is defined by its measurement: time is what a clock reads. In classical, non-relativistic physics it is a scalar quantity and, like length, mass, and charge, is usually described as a fundamental quantity. Time can be combined mathematically with other physical quantities to derive other concepts such as motion, kinetic energy and time-dependent fields.

Time is used to define other quantities such as velocity, so defining time in terms of such quantities would result in circularity of definition. An operational definition of time, wherein one says that observing a certain number of repetitions of one or another standard cyclical event (such as the passage of a free-swinging pendulum) constitutes one standard unit such as the second, is highly useful in the conduct of both advanced experiments and everyday affairs of life. The operational definition leaves aside the question whether there is something called time, apart from the counting activity just mentioned, that flows and that can be measured.

### 3.2 Newton's Views on Space and Time

Isaac Newton founded classical mechanics on the view that space is distinct from body and that time passes uniformly without regard to whether anything happens in the world. For this reason he spoke of absolute space and absolute time, so as to distinguish these entities from the various ways by which we measure them (which he called relative spaces and relative times). From antiquity into the eighteenth century, contrary views which denied that space and time are real entities maintained that the world is necessarily a material plenum. Concerning space, they held that the idea of empty space is a conceptual impossibility. Space is nothing but an abstraction we use to compare different arrangements of the bodies constituting the plenum. Concerning time, they insisted, there

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can be no lapse of time without change occurring somewhere. Time is merely a measure of cycles of change within the world.

### 3.3 Frame of Reference

A frame of reference is a set of coordinates (for example x, y & z axes) with respect to which any physical quantity such as velocities and positions can be determined.

Figure 2. 1 A frame of Reference

### 3.4 Inertial Frames of Reference

The inertia of a body is the resistance the body offers to changing its state of motion. - Uniformly moving reference frames (e.g. those considered at 'rest' or moving with constant velocity in a straight line) are called inertial reference frames. They are frames

in which Newton's Laws are valid.

Special relativity deals only with physics viewed from inertial reference frames. - If we can neglect the effect of the earth's rotations, a frame of reference fixed in the earth is an inertial reference frame. An example of a non-inertial frame is a rotating frame, such as a carousel.

Einstein's first postulate, called the principle of relativity, states: The laws of physics are the same in every inertial frame of reference. If the laws differ, that difference could distinguish one inertial frame from the others or make one frame somehow more "correct" than another. Here are two examples. Suppose you watch two children playing catch with a ball while the three of you are aboard a train moving with constant velocity. Your observations of the motion of the ball, no matter how carefully done, can't tell you how fast (or whether) the train is moving. This is because Newton's laws of motion are the same in every inertial frame. Another example is the electromotive force (emf) induced in a coil of wire by a nearby moving permanent magnet. In the frame of reference

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in which the coil is stationary (Fig. 2.2a), the moving magnet causes a change of magnetic flux through the coil, and this induces an emf. In a different frame of reference in which the magnet is stationary (Fig. 2.2b), the motion of the coil through a magnetic field induces the emf. According to the principle of relativity, both of these frames of reference are equally valid. Hence the same emf must be induced in both situations shown in Fig. 2.2. Indeed, all of the laws of electromagnetism are the same in every inertial frame of reference.

Figure 2.2: The same emf is induced in the coil whether (a) the magnet moves relative to the coil or (b) the coil moves relative to the magnet.

During the 19th century, most physicists believed that light traveled through a hypothetical medium called the ether, just as sound waves travel through air. If so, the speed of light measured by observers would depend on their motion relative to the ether and would therefore be different in different directions. The Michelson-Morley experiment was an effort to detect motion of the earth relative to the ether. Einstein's conceptual leap was to recognize that if Maxwell's equations are valid in all inertial frames, then the speed of light in vacuum should also be the same in all frames and in all directions.

#### Self-Assessment Exercise(s) 1

1. Briefly define frames of references
2. What do you understand by Inertial Frame of References?

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#### 4.0 Conclusion

This unit took you through the basic concept of space, time and frame of reference, we were able to see how Newton understudied space and time and how Einstein's postulate helped us understand Frame of reference.

## 5.0 Summary

All of the fundamental laws of physics have the same form in all inertial frames of reference. The speed of light in vacuum is the same in all inertial frames and is independent of the motion of the source. Simultaneity is not an absolute concept; events that are simultaneous in one frame are not necessarily simultaneous in a second frame moving relative to the first.

## 6.0 Tutor Marked Assignment

1. Explain in clear term the principle of space and time.
2. Define frame of reference?
3. Describe Einstein's postulates of frame of reference.

## 7.0 References/Further Readings

Fundamentals of physics (10th Edition) Jearl Walker, 2014

University physics with modern physics (13th Edition) Hugh D. Young and Roger A. Freedman, 2012.

Worked Examples from Introductory Physics. Volume I: David Murdock, 2005.

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PHY113

General Physics

I

Module 2

Motion and Forces

Unit 1: Scalars and Vectors

Unit 2: Velocity and acceleration

Unit 3: Newton's laws of motion

Unit 4: Collision, momentum and Impulse

Unit 5: Work, Energy and Power

Unit 1

Scalars and Vectors

Content

1.0 Introduction

2.0 Learning Outcomes

3.0 Learning Content

3.1 Definition of scalar and vector quantities

3.2 The difference between scalars and vectors

3.3 Vector addition and subtraction

3.4 Relative velocity

4.0 Conclusion

5.0 Summary

6.0 Tutor-Marked Assignment

7.0 References/Further Reading

## 1.0 Introduction

Some physical quantities, such as time, temperature, mass, and density, can be described completely by a single number with a unit. But many other important quantities in physics have a direction associated with them and cannot be described by a single number. A simple example is describing the motion of an airplane: We must say not only how fast the plane is moving but also in what direction. The speed of an object combined with its direction of motion is an important concept worth considering in physics.

## 2.0 Learning Outcomes

At the end of this unit, you should be able to:

- i. Explain the concept of scalars and vectors
- ii. Differentiate between scalar and vector quantities and their representation
- iii. Conceptualize vector addition and subtraction operation
- iv. Conceptualize relative velocity

## 3.0 Learning Content

### 3.1 Definition of Scalar and Vector Quantities

#### 3.1.1 Scalar Quantities

A quantity which is completely specified by a certain number associated with a suitable unit without any mention of direction in space is known as scalar. Examples of scalar are time, mass, length, volume, density, temperature, energy, distance, speed etc. The number describing the quantity of a particular scalar is known as its magnitude. The scalars are added subtracted, multiplied and divided by the usual arithmetical laws.

For example,  $6 \text{ kg} + 3 \text{ kg} = 9 \text{ kg}$ ,

$4 \times 2 \text{ s} = 8 \text{ s}$ .

#### 3.1.2 Vector Quantities

A quantity which is completely described only when both their magnitude and direction are specified is known as vector. In contrast, a vector quantity has both magnitude (the “how much” or “how big” part) and direction in space. Examples of vectors are force, velocity, acceleration, displacement, torque, momentum, gravitational force, electric and magnetic intensities etc. Combining vectors requires a different set of operations. To understand more about vectors and how they combine, we start with the simplest vector quantity

Vector Representations:

A vector quantity is represented by a straight line segment, say  $\vec{PQ}$

PQ. The arrow head

indicates the direction from P to Q. The length of the vector represents its magnitude.

Sometimes the vectors are represented by single letter such as  $\vec{V}$  or  $\vec{V}$ . The magnitude

$|\vec{V}|$  means modulus of  $\vec{V}$

of a vector is denoted by  $|\vec{V}|$  or by just  $V$ , where  $|\vec{V}|$  which is a positive value

## Types of Vectors

### Unit vector

A vector whose magnitude is unity i.e., 1 and directed along the given vector is called a unit vector. If  $\vec{a}$  is a vector then a unit vector in the direction of  $\vec{a}$ , denoted by  $\hat{a}$  (read as a cap) is given as,

$$\hat{a} =$$

$$\frac{\vec{a}}{|\vec{a}|}$$

$$|\vec{a}|$$

Where  $|\vec{a}|$  is the magnitude of vector  $\vec{a}$

### Free Vector

A vector whose position is not fixed in space. Thus, the line of action of a free vector can be shifted parallel to itself. Displacement is an example of a free vector as shown

**Localized or Bounded Vectors:** A vector which cannot be shifted parallel to itself, i.e., whose line of action is fixed is called a localized or bounded vector. Force and momentum are examples of localized vectors.

**Coplanar Vectors:** The vectors which lie in the same plane are called coplanar vectors

**Concurrent Vectors:** The vectors which pass through the common point are called concurrent vectors

**Negative of a Vector:** The vector which has the same magnitude as the vector " $\vec{a}$ " but opposite in direction to  $\vec{a}$  is called the negative to " $\vec{a}$ "

**Null or Zero Vector:** It is a vector whose magnitude is zero. We denote the null vector by  $\vec{0}$ . The direction of a zero vector is arbitrary.

**Equal Vectors:** Two vectors  $\vec{a}$  and  $\vec{b}$  are said to be equal if they have the same magnitude and direction. If  $\vec{a}$  and  $\vec{b}$  are equal vectors then  $\vec{a} = \vec{b}$

## 3.2 Addition and Subtraction of Vectors

### Addition of Vectors

Suppose  $\vec{a}$  and  $\vec{b}$  are any two vectors. Choose point A so that  $\vec{a} = \vec{OA}$

and choose point

C so that  $\vec{b} = \vec{AC}$

The sum  $\vec{a} + \vec{b}$  of  $\vec{a}$  and  $\vec{b}$  is the vector  $\vec{OC}$

Thus the sum of two

vectors  $\vec{a}$  and  $\vec{b}$  is performed by the Triangle Law of addition

### Subtraction of Vectors

If a vector  $\vec{b}$  is to be subtracted from a vector  $\vec{a}$ , the difference, vector  $\vec{a} - \vec{b}$  can be obtained by adding vectors  $\vec{a}$  and  $-\vec{b}$ . The vector  $-\vec{b}$  is a vector which is equal and parallel to that vector  $\vec{b}$  but its arrow-head points in opposite direction. Now the  $-\vec{b}$  can be added by the head-to-tail rule. Thus the line

segment AC represents, in

magnitude and direction, the vector  $\vec{a} - \vec{b}$

### Properties of Vector Addition

#### 1. Vector addition is Commutative

i.e.  $\vec{a} + \vec{b} = \vec{b} + \vec{a}$  where  $\vec{a}$  and  $\vec{b}$  are any two vectors

#### 2. Vector addition is Associative

i.e.  $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$  where  $\vec{a}, \vec{b}$  and  $\vec{c}$  are any three vectors

$\vec{0}$  is the identity in vectors addition

i.e.  $\vec{a} + \vec{0} = \vec{a}$

i.e. for every vector  $\vec{a}$

$\vec{a} + \vec{0} = \vec{a}$  Where  $\vec{0}$  is the zero vector

$\vec{a} + \vec{0} = \vec{a}$  Where  $\vec{0}$  is the zero vector

### Self-Assessment Exercise(s) 1

1. Which of the following is not a vector quantity ? A. Speed B. Velocity C. Force

D. Acceleration E. Electric Field

2. Which of the following is not a scalar quantity? A. Density B. Weight C. Speed

D. Mass E. Temperature

3. Two forces, whose resultant is 100N, are perpendicular to each other. If one of them makes an angle of  $60^\circ$  with the resultant, calculate its magnitude: (Sin  $60^\circ = 0.8660$ , Cos  $60^\circ = 0.5000$ )

A. 200. N B. 173.2N C. 115.5N D. 86.6N

4. Distinguish between scalar and vector quantities. Which of the following are

scalars and which are vectors? Speed, velocity, mass, density, force, displacement, weight, acceleration, momentum

5. A toy train moves along a winding track at an average speed of 0.25m/s. How far will it travel in 4.00 minutes?

### 4.0 Conclusion

In this unit you have learnt the concept of scalar and vector quantities, types of vectors and how vectors addition and subtraction operation are carried out.

### 5.0 Summary

In this unit, we have defined scalar quantities as numbers and can combine with the usual rules of arithmetic. Vector quantities have direction as well as magnitude and combined according to the rules of vector algebra. The negative of a vector has the same magnitude but points in the opposite direction.

### 6.0 Tutor-Marked Assignments

1. What do you understand by the word scalar and vector quantities?

2. Explain the meaning of resultant of two vectors?

3. What are the types of vectors?

### 7.0 References/Further Reading

Hannagan, T.J. (1982) College Physics, Macmillan Press Ltd.

M.W. Anyakoha (Ph.D) New School Physics for Senior Secondary Schools, (1991)

Scalar and Vectors.

## Unit 2

## Velocity and Acceleration

## Content

## 1.0 Introduction

## 2.0 Learning Outcomes

## 3.0 Learning Content

3.1 Understanding of physical quantities such as velocity and acceleration concepts

3.2 How to determine the vector velocity of a body from a knowledge of its path.

3.2 How to find the vector acceleration of a body, and why a body can have an acceleration even if its speed is constant.

## 4.0 Conclusion

## 5.0 Summary

## 6.0 Tutor-Marked Assignment

## 7.0 References/Further Reading

## 1.0 Introduction

In this unit we will study velocity and acceleration, the part of mechanics that enables us to describe motion. We will concentrate on the simplest kind of motion: a body moving along a straight line. To describe this motion, we introduce the physical quantities velocity and acceleration. An important note is that both velocity and acceleration are vectors. We'll develop simple equations to describe straight-line motion in the important special case when the acceleration is constant.

## 2.0 Learning Outcomes

At the end of this unit, you should be able to:

- i. describe straight-line motion in terms of average velocity, instantaneous velocity, average acceleration, and instantaneous acceleration.
- ii. interpret graphs of position versus time, velocity versus time, and acceleration versus time for straight-line motion.
- iii. solve problems involving straight-line motion with constant acceleration, including free-fall problems.

## 3.0 Learning Content

## 3.1 Velocity

## 3.1.1 Definition of Velocity

Velocity is a vector quantity that embraces both the speed and the direction of motion.

If an object undergoes displacement  $\Delta x$  in a time interval  $t$ , then

average velocity

average velocity =

displacement / time

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} \dots\dots\dots (1)$$

□ □

The direction of the velocity vector is the same as that of the displacement vector. The SI unit of velocity (and speed) are those of distance divided by time such as m/s other units include cm/sec, km/h

### Instantaneous Velocity

Sometimes the average velocity is all you need to know about a particle's motion but the average velocity of a particle during a time interval can't tell us how fast or in what direction, the particle is moving at any given time during the interval. To do this we need to know the velocity at any specific instant of time or specific point along the path. This is called instantaneous velocity, and it need to be defined carefully.

### Definition of Instantaneous Velocity

The instantaneous velocity is the limit of the average velocity as the time interval approaches zero; it equals the instantaneous rate of change of position with time.

Instantaneous velocity, like average velocity is a vector quantity

As the average velocity evaluated for a time interval that approaches zero, instantaneous velocity can be determined. Thus, if an object undergoes a displacement  $\Delta x$  in a time  $\Delta t$ , then for that object the instantaneous velocity is

$\Delta x$  □ □

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \dots\dots\dots(2)$$

$\Delta t \rightarrow 0$

### Uniformly Accelerated Motion

Considering a particle moving along the x-axis. Suppose that at time  $t_1$  the particle is at point P1 and has x-component of (instantaneous velocity)  $v_{1x}$ , and at a later time  $t_2$  it is at point P2 and has x-component of velocity  $v_{2x}$ . so the x-component of velocity changes by an amount  $\Delta v_x = v_{2x} - v_{1x}$  during the time interval  $\Delta t = t_2 - t_1$ .

Therefore, acceleration measures the time rate-of-change of velocity along the component of any predefined direction:

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= (□□□□□□□ □ - □□□□□□□□□□□□□□, ).....(3)

□ -□ 2 1 □□

Where  $v_{2x}$  = final velocity,  $v_{1x}$  = initial velocity, and

$(t_2 - t_1)$  is the time interval over which the change occurred.

The units of acceleration are those of velocity divided by time is (m/s<sup>2</sup>) or km/hr<sup>2</sup>

### Instantaneous Acceleration

We can now define instantaneous acceleration following the same procedure that we used to define instantaneous velocity.

### 3.1.2 Definition of Instantaneous Acceleration

The instantaneous acceleration is the limit of the average acceleration as the time interval approaches zero. In the language of calculus, instantaneous acceleration equals the instantaneous rate of change of velocity with time

Thus

$$\Delta v = \Delta x / \Delta t$$

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = (\text{instantaneous x-acceleration}) \dots\dots\dots(4)$$

$$\Delta t \rightarrow 0 \quad \Delta x \rightarrow \Delta x$$

Uniform accelerated motion along a straight line:

This is an important situation. In this case, the acceleration vector is constant and lies along the line of the displacement vector, so that the direction of  $\Delta x$  and  $\Delta v$  can be specified with plus and minus signs. Direction is important, and a positive direction must be chosen when analyzing motion along a line. Either direction may be chosen as positive. If a displacement, velocity or acceleration is in the opposite direction, it must be taken as negative.

### 3.2 Proof of equations of motion

If we represent the displacement by  $x$  (positive if in the positive direction, and negative if in the negative direction), then the motion can be described with the five equations for uniformly accelerated motion

Assuming the x-acceleration  $a$  is constant, the average x-acceleration  $\frac{v_2 - v_1}{t_2 - t_1}$  for any time interval is the same as  $a$ . Therefore the derivation of equations of motion for the position  $x$  and the x-velocity  $v_x$  as functions of time is given by.

To find an expression for  $v_x$  we first replace  $\frac{v_2 - v_1}{t_2 - t_1}$  by  $a$  :

$$v_2 - v_1 = a(t_2 - t_1)$$

$$v_2 = v_1 + a(t_2 - t_1)$$

$$\dots\dots\dots(5)$$

$$v_2 = v_1 + a t$$

Now we let  $t_1 = 0$  and let  $t_2$  be any later time  $t$ . we use the symbol  $v_0$  for the x- velocity at the initial time  $t=0$ ; the x-velocity at the later time  $t$  is  $v$  . Then we have

$$v - v_0 = a t$$

$$v = v_0 + a t \dots\dots\dots(6)$$

$$x = v_0 t + \frac{1}{2} a t^2 \quad (\text{constant x- acceleration only}) \dots\dots\dots(7)$$

Next we'll derive an equation for the position  $x$  as a function of time when the x-acceleration is constant. To do this, we use two different expressions for the average x-velocity  $\frac{v_2 - v_1}{t_2 - t_1}$  during the interval from  $t = 0$  to any later time  $t$ . The first expression comes from the definition of  $\frac{v_2 - v_1}{t_2 - t_1}$  , which is true whether or not the acceleration is constant. We call the position at time the initial position, denoted by  $x_0$  . The position at the later time  $t$  is simply  $x$ . Thus for the time interval the average velocity gives

$$\frac{x - x_0}{t - 0} = \frac{v_2 - v_1}{t_2 - t_1}$$

$$\frac{x - x_0}{t} = \dots\dots\dots(8)$$

□

We can also get a second expression for  $x - x_0$  that is valid only when the x-acceleration is constant, so that the x-velocity changes at a constant rate. In this case the average x-velocity for the time interval from 0 to t is simply the average of the x-velocities at the beginning and end of the interval:

□ + □

$$x - x_0 = \frac{0 + v}{2} t \quad (\text{constant } x\text{-acceleration only}) \quad \dots\dots\dots(9)$$

(This equation is not true if the x-acceleration varies during the time interval.)

We also know that with constant x-acceleration, the x-velocity  $v$  at any time t is given by Equation (1) Substituting that expression for  $v$  into Equation (2), we find

$$x - x_0 = \frac{0 + v}{2} t = \frac{0 + v_0 + at}{2} t$$

$$x - x_0 = \frac{0 + v_0}{2} t + \frac{at^2}{2} \quad \dots\dots\dots(10)$$

2

□□ □

$$x - x_0 = v_0 t + \frac{1}{2} at^2 \quad (\text{constant } x\text{-acceleration only}) \quad \dots\dots\dots(11)$$

2

Finally, we set Eqs. (2.9) and (2.11) equal to each other and simplify:

$$v_0 t + \frac{1}{2} at^2 = \frac{v_0 + v}{2} t$$

$$\frac{1}{2} at^2 = \frac{v - v_0}{2} t \quad \dots\dots\dots(12)$$

2 □

$$at^2 = v - v_0$$

$$v = v_0 + at \quad (\text{constant } x\text{-acceleration only}) \quad \dots\dots\dots(13)$$

2

$$\text{Make } t \text{ the subject of the formula in equation (1) - } v = v_0 + at \quad \dots\dots\dots(14)$$

$$v - v_0 = at$$

$$t = \frac{v - v_0}{a} \quad \dots\dots\dots(15)$$

□□

$$x - x_0 = v_0 t + \frac{1}{2} at^2$$

$$x - x_0 = v_0 \left( \frac{v - v_0}{a} \right) + \frac{1}{2} a \left( \frac{v - v_0}{a} \right)^2$$

□□

$$x - x_0 = \frac{v_0(v - v_0)}{a} + \frac{1}{2} \frac{(v - v_0)^2}{a} \quad \dots\dots\dots(16)$$

□□ 2

Simplifying further gives us

2

$$x - x_0 = \frac{v_0(v - v_0)}{a} + \frac{1}{2} \frac{(v - v_0)^2}{a} \quad (\text{constant } x\text{-acceleration only}) \quad \dots\dots\dots(17)$$

are the equations of motion with constant acceleration. By using these equations, we can solve any problem involving straight-line motion of a particle with constant acceleration.

### 3.3 Relative Velocity

You've no doubt observed how a car that is moving slowly forward appears to be

moving backward when you pass it. In general, when two observers measure the velocity of a moving body, they get different results if one observer is moving relative to the other. The velocity seen by a particular observer is called the velocity relative to that observer, or simply relative velocity.

#### Self-Assessment Exercise(s) 1

1. A bus moving in a straight line at a speed of 20m/s begins to slow at a constant rate of 3.0m/s each second. Find how far it goes before stopping ?
2. a body with initial velocity 8.0m/s moves along a straight line with a constant acceleration and travels 640m in 40s. For the 40s interval, find a. average velocity b. the final velocity c. the acceleration

#### 4.0 Conclusion

In this unit you should have learnt that distance and speed are scalar quantities while velocity and acceleration are vector quantities. Also the explain the terms used in the equations of motion

#### 5.0 Summary

- 1 Displacement is defined as the distance travelled in a specified direction. It is a vector quantity.
2. Speed is the distance travelled in unit time. It is a scalar quantity
3. Velocity is the rate of change of displacement
4. Acceleration is the rate of change of velocity with time. It is a vector quantity
3. Equations of uniformly accelerated motions are;

$$v = u + at$$

$$v^2 = u^2 + 2as$$

1

$$s = ut + \frac{1}{2}at^2$$

2

1

$$s = \left( \frac{u+v}{2} \right)t$$

2

#### 6.0 Tutor-Marked Assignments

1. What do you understand by the word displacement, speed, velocity, instantaneous velocity, acceleration, instantaneous acceleration?
2. A body moving with uniform acceleration  $a$  has two points (5, 15) and (20, 60) on the velocity – time graph of its motion. Calculate  $a$  A. 0.25m/s<sup>2</sup> B. 3.00m/s<sup>2</sup> C. 4.00m/s<sup>2</sup> D. 9.00m/s<sup>2</sup> E. 16.00m/s<sup>2</sup>
3. Distinguish between uniform acceleration and variable acceleration. Illustrate by means of a velocity time graph

#### 7.0 References/Further Reading

M.W. Anyakoha (Ph.D )New School Physics for Senior Secondary Schools, (1991)  
Scalar and Vectors.

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## Unit 3

### Newton's Law of Motion

#### Content

##### 1.0 Introduction

##### 2.0 Learning Outcomes

##### 3.0 Learning Content

##### 3.1 Mass and Weight

##### 3.2 Force and the Newton

##### 4.0 Conclusion

##### 5.0 Summary

##### 6.0 Tutor-Marked Assignment (TMA)

##### 7.0 References/Further Reading

##### 1.0 Introduction

Motion has been discussed in terms of displacement, velocity and acceleration. Now we want to discuss about why does a body remain at rest, and a body moving with uniform velocity continue its uniform motion? What will make the body at rest to move, and a body moving with uniform velocity to change its state of motion? Such an agent that changes or tends to change the state of rest or of uniform motion in a straight line of a body is called Force

##### 2.0 Learning Outcomes

At the end of this unit, you should be able to explain:

- i. the concept of mass and weight in physics
- ii. The significance of net force on an object, and what happens when the net force is zero.
- iii. The relationship among the net force on an object, the object's mass, and its acceleration.
- iv. Newton's laws of motion

##### 3.0 Learning Content

##### 3.1 Mass and Weight

One of the most familiar forces is the weight of a body, which is the gravitational force that the earth exerts on the body. (If you are on another planet, your weight is the gravitational force that planet exerts on you.) Unfortunately, the terms mass and weight are often misused and interchanged in everyday conversation. It is absolutely essential for you to understand clearly the distinctions between these two physical quantities.

##### 3.1.1 Mass

Mass is a property of an object which is one way of specifying the quantity of matter in it. The concept of mass can be explained formally in terms of the effect of a total force on a body, using the equation of motion discussed later in this chapter. Alternatively,

mass can be determined in terms of the gravitational force on the object. At a fixed location the gravitational force on different objects (their weights) is proportional to their masses. The SI unit of mass is the kilogram (symbol kg) and the unit of weight is the newton (N). Mass characterizes the inertial properties of a body. The greater the mass, the greater the force needed to cause a given acceleration; this is reflected in Newton's second law,

### 3.1.2 Weight

Weight, on the other hand, is a force exerted on a body by the pull of the earth. The force that makes the body accelerate downward is its weight. Any body near the surface of the earth that has a mass of 1 kg must have a weight of 9.8 N to give it the acceleration we observe when it is in free fall.

More generally, a body with mass  $m$  must have weight with magnitude  $w$  given by  $W = mg$  (magnitude of the weight of a body of mass  $m$ ).....(1)  
 $g$  is the magnitude of the acceleration due to gravity

Hence the magnitude  $w$  of a body's weight is directly proportional to its mass  $m$  and its always positive.

### 3.1.3 Similarities between Mass and Weight

Mass and weight are related. Bodies having large mass also have large weight. A large stone is hard to throw because of its large mass, and hard to lift off the ground because of its large weight.

To understand the relationship between mass and weight, note that a freely falling body has an acceleration of magnitude  $g$ , called the acceleration due to gravity.

Newton's second law tells us that a force must act to produce this acceleration. If a 1-kg body falls with an acceleration due to gravity the required force has magnitude.

$$F = ma \text{ (1kg) (9.8m/s}^2\text{)} = 9.8\text{N}.....(2)$$

## 3.2 Newton's Laws of Motion

How do the forces that act on a body affect its motion? To begin to answer this question, you would almost certainly agree that if a body is at rest, and if no net force acts on it (that is, no net push or pull), that body will remain at rest. The tendency of a body to keep moving once it is set in motion results from a property called inertia.

### 3.2.1 Newton First Law of Motion

This tendency of bodies to remain in their state of rest or uniform linear motion in the absence of applied forces is known as inertia. Newton's First Law shows that inertia is inherent in a body at rest or the one moving with a constant velocity. Inertia is a property of matter. Mass is a measure of inertia, the more massive an object is, the more inertia it has.

Newton's first law explains why passengers in a fast moving vehicle tend to move forward when a car suddenly stops, or to be pushed backwards when the vehicle suddenly speeds off, since there is little or no force to restrain them. Hence safety belts are used to provide this restraining force.

Newton's first law also explains what force does not suggest how force should be measured.

#### Newton's First Law

A body acted on by no net force moves with constant velocity (which may be zero) and zero acceleration. Or an object at rest will remain at rest; an object in motion will continue in motion with constant velocity, except insofar as it is acted upon by an external force. Force is the changer of motion

#### 3.2.2 Newton Second Law of Motion

Newton's second law is a fundamental law of nature which express the basic relationship between force and motion. Newton's second law enables us to define an absolute unit of force which remains constant under all conditions. In symbols the law states that

$\Delta p = F \Delta t$

or

$F = \frac{\Delta p}{\Delta t}$

or

$F = \frac{mv - mu}{t - u} \dots \dots \dots (3)$

Where

Where m, u, v, t are the mass, initial velocity, final velocity and time respectively of motion of the body acted upon by a force F; and the product of m and v is called the momentum

Hence

or

$F = \frac{\Delta p}{\Delta t}$

$F = \frac{mv - mu}{t - u} \dots \dots \dots (4)$

or

Where = acceleration, a.

or

Thus  $F = kma \dots \dots \dots (5)$

The unit of force is chosen so that  $k = 1$ . Hence we can write

$F = ma \dots \dots \dots (6)$

Also,

or

$F = \frac{\Delta p}{\Delta t}$

Or  $F \cdot t = \Delta p \dots \dots \dots (7)$

The product  $F \cdot t$  is the impulse, I of the force

$I = Ft = \text{change in momentum} \dots \dots \dots (8)$

Example

An unbalanced force of 20N acts on a 4.0kg mass. What acceleration does it give it ?

Solution

$$v = u$$

$$20 = 4v$$

$$v = 5\text{ m/s}$$

Example

A ball of mass 0.3kg, moving at a velocity of 20m/s is suddenly hit by a force of 5N for a time of 0.03 sec. Find its new velocity of motion.

Solution

From Newton's second law

$$F = ma$$

$$F = m(v - u)$$

$$\text{Impulse } F \cdot t = m(v - u)$$

$$5 \cdot 0.03$$

$$= 0.3 - 20$$

$$v$$

$$V = 20.5\text{ m/s}$$

If a net external force acts on a body, the body accelerates. The direction of acceleration is the same as the direction of the net force. The mass of the body times the acceleration of the body equals the net force vector.

$$\Sigma F = ma \dots\dots\dots(9)$$

An alternative statement is that the acceleration of a body is in the same direction as the net force acting on the body, and is equal to the net force divided by the body's mass:

$$\Sigma F$$

$$a = \dots\dots\dots(10)$$

$$a$$

The acceleration  $a$  has the same direction as the resultant force  $F$ .

### 3.2.3 Newton's Third Law

A force acting on a body is always the result of its interaction with another body, so forces always come in pairs. You can't pull on a doorknob without the doorknob pulling back on you. When you kick a football, the forward force that your foot exerts on the ball launches it into its trajectory, but you also feel the force the ball exerts back on your foot. If you kick a boulder, the pain you feel is due to the force that the boulder exerts on your foot. In each of these cases, the force that you exert on the other body is in the opposite direction to the force that body exerts on you. Experiments show that whenever two bodies interact, the two forces that they exert on each other are always equal in magnitude and opposite in direction. This fact is called Newton's third law of motion:

#### Newton's Third Law

If body A exerts a force on body B (an "action"), then body B exerts a force on body A (a "reaction"). These two forces have the same magnitude but are opposite in direction.

These two forces act on different bodies.

#### Self-Assessment Exercise(s) 1

1. A worker applies a constant horizontal force with magnitude 20 N to a box with mass 40 kg resting on a level floor with negligible friction. What is the acceleration of the box?
2. State Newton's laws of motion.

#### 4.0 Conclusion

In this unit, you have learnt the definition of mass and weight, similarities of mass and weight. In addition, force, motion and Newton's laws of motion

#### 5.0 Summary

Mass is the quantity of matter in a body. It is also a measure of the inertia of a body.

Mass is constant all over the earth.

Weight: The weight of a body is the force with which the earth pulls the body to the earth's centre. Weight varies from place to place, because the  $g$  in  $F=mg$  varies from place to place.

#### Newton's laws of motion

First Law: Every object continues in its state of rest or uniform motion in a straight line unless acted upon by an external force. From this law, we obtain the idea of inertia.

Second Law: The rate of change of momentum of a body is proportional to the impressed force and takes place in the direction of the force

Third law states that action and reaction are equal and opposite.

#### 6.0 Tutor-Marked Assignments

1. A body of mass 0.1kg dropped from a height of 8m onto a hard floor bounces back to a height of 2m. Calculate the change of momentum. If the body is in the contact with the floor for 0.1sec. What is the force exerted on the body? ( $g=10\text{m/s}^2$ )
2. A body of mass 2kg falls freely from rest through a height of 50m and comes to rest having penetrated 5.0cm of sand. Calculate (i) the velocity with which the body hits the sand, (ii) the time taken in falling (iii) the average force exerted by the sand in bringing the body to rest.

#### 7.0 References/Further Reading

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M.W. Anyakoha (Ph.D) New School Physics for Senior Secondary Schools, (1991)

Scalar and Vectors.

#### Unit 4

Collision, Momentum  
and Impulse

#### Content

##### 1.0 Introduction

##### 2.0 Learning Outcomes

##### 3.0 Learning Content

## 3.1. Collision Theory

## 3.2 Momentum and Impulse

## 4.0 Conclusion

## 5.0 Summary

## 6.0 Tutor-Marked Assignment (TMA)

## 7.0 References/Further Reading

## 1.0 Introduction

There are many questions involving forces that cannot be answered by directly applying Newton's second law, for example, when a moving van collides head-on with a compact car, what determines which way the wreckage moves after the collision?

Our approach uses two new concepts, momentum and impulse, and a new conservation law, conservation of momentum. This conservation law is every bit as important as the law of conservation of energy. The law of conservation of momentum is valid even in situations in which Newton's laws are inadequate

## 2.0 Learning Outcomes

By the end of this unit, you should be able to:

- i. Explain collision Theory
- ii. Solve problems in which two bodies collide with each other
- iii. Briefly explain the important distinction among elastic and inelastic collisions.
- iv. State and explain the meaning of the law of conservation of linear momentum

## 3.0 Learning Content

## 3.1 Collision Theory

When we talk about a collision in physics (between two particles, say) we mean that two particles are moving freely through space until they get close to one another; then, for a short period of time they exert strong forces on each other until they move apart and are again moving freely.

There are two principal types of collision – elastic and inelastic collisions.

If in a collision between two bodies, the total kinetic energy is the same before and after the collisions, the collision is said to be completely or perfectly elastic. Thus in such an elastic collision both the kinetic energy and momentum are conserved.

Let us consider two objects of masses  $m_1$  and  $m_2$  moving with initial velocities  $u_1$  and  $u_2$  before collision and with final velocities  $v_1$  and  $v_2$  after collision in the same direction. If the collision is perfectly elastic, we can write two equations from the law of conservation of momentum and the law of conservation of kinetic energy. Hence,

$$(m_1 u_1 + m_2 u_2) = (m_1 v_1 + m_2 v_2) \dots \dots \dots (1)$$

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \dots \dots \dots (2)$$

Two examples of collisions that are often very nearly perfectly elastic collisions are the collision of billiard balls and of molecules and atoms. Also a tennis ball or ping – pong ball that bounces back to its original height after a surface is said to have had an elastic

collision with the surface. In a perfectly elastic head-on collision between two bodies, the relative velocity of the two bodies is unchanged in magnitude but reversed in direction.

In an inelastic collision. The kinetic energy decreases after collision but the momentum is still conserved. In inelastic collision, the colliding bodies stuck together and move as a unit after collision. This means that the velocities of the two bodies after collision are

$$v_1 = v_2 = v \dots \dots \dots (3)$$

From conservation of linear momentum we have

$$(m_1 v_1 + m_2 v_2) = (m_1 v_1 + m_2 v_2) = (m_1 + m_2) v \dots \dots \dots (4)$$

The kinetic energy of the system before collision is given by

$$E_{K1} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \dots \dots \dots (5)$$

And after collision, the kinetic energy is

$$E_{K2} = \frac{1}{2} (m_1 + m_2) v^2 \dots \dots \dots (6)$$

For a completely elastic collision, the kinetic energy before collision is greater than the kinetic energy after collision i.e  $E_{K1} > E_{K2}$

The sum of the momenta before and after the collision is conserved. While the total momentum is conserved for a system of isolated colliding particles, the mechanical energy may or may not be conserved. If the mechanical energy (usually meaning the total kinetic energy) is the same before and after a collision, we say that the collision is elastic. Otherwise we say the collision is inelastic. If two objects collide, stick together, and move off as a combined mass, we call this a perfectly inelastic collision.

When a ball is dropped to the ground, one of four things may happen:

1. It may rebound with exactly the same speed as the speed at which it hit the ground. This is an elastic collision.
2. It may come to a complete rest, for example if it were a ball of soft putty. I shall call this a completely inelastic collision.
3. It may bounce back, but with a reduced speed. For want of a better term I shall refer to this as a somewhat inelastic collision.
4. If there happens to be a little heap of gunpowder lying on the table where the ball hits it, it may bounce back with a faster speed than it had immediately before collision. That would be a super elastic collision.

Let us consider the case of a ball hitting a surface.

$$e = \frac{\text{relative speed after collision}}{\text{relative speed before collision}} \dots \dots \dots (7)$$

is called the coefficient of restitution, for which I shall use the symbol  $e$ .

The coefficient is 1 for an elastic collision, less than 1 for an inelastic collision, zero for a completely inelastic collision, and greater than 1 for a super elastic collision.

### 3.2 Momentum

Any moving object can have momentum. This is because momentum is mass in motion. The way we determine an object's momentum is fairly straightforward.

Momentum is the object's mass times its velocity.

in equation form,  $p = mv$ , .....(8)

where p is momentum,

m is mass in kilograms, and v is velocity in meters per second.

Momentum is proportional to both mass and velocity, meaning that a change in one will cause the same amount of change in the other. So if you increase an object's mass, you also increase its momentum. The same is true for velocity: increase or decrease the object's speed, and you increase or decrease its momentum by the same amount. But usually it's the object's velocity that changes instead of its mass, right? You may remember that a change in velocity means the object is accelerating. You may also remember that acceleration is caused by a force and that the greater the force, the greater the acceleration. Therefore, the greater the acceleration, the greater the momentum!

Example:

A body of mass 3.0 kg moves with a velocity of 10m/s. Calculate the momentum of the body.

Solution

$$p = mv$$

$$3 \times 10 = 30\text{kgm/s}$$

### 3.3 Impulse

Impulse is usually associated with collision. When a body collides with another, each receives an impulse or blow. The impulse consists of a large varying force acting for a very short time. This change in momentum caused is called impulse. The greater the impulse, the greater the change in momentum. To change the impulse, you can either change the amount of force, or you can change the time interval in which that force acts. In equation form, we can write this relationship between impulse and momentum as:

#### 3.3.1 Definition of impulse

We define impulse as the product of the average force acting on a particle and the time during which it acts.

If a force F acts for a short time t, the impulse I, is given by:

$$I = Ft$$

$$F = \frac{\Delta p}{\Delta t} \Rightarrow Ft = \Delta p$$

$$Ft = mv = I = \text{Impulse}$$

$$I = F \times t \text{ ..... (9)}$$

The unit of impulse is the Newton – second. Impulse is a vector. It has the same direction as the direction of force.

Example: A stationary ball is hit by an average force 50N for a time of 0.03sec. What is the impulse experienced by the body?

Solution

$$I = Ft$$

$$= 50 \times 0.03$$

$$= 1.5 \text{ N-sec}$$

### 3.4 Conservation of Linear Momentum

The principle of the conservation of linear momentum can be stated in different ways:

1. In any system of colliding objects the total momentum is always conserved provided that there is no net external force acting on the system OR
2. The total momentum of an isolated or closed system of colliding bodies remains constant
3. If two or more bodies collide in a closed system, the total momentum after the collision is equal to the total momentum before the collision

The principle of the conservation of linear momentum follows from Newton's third and second laws of motion.

Consider two bodies A and B of masses  $m_A$  and  $m_B$  moving towards each other with velocities  $u_A$  and  $u_B$  respectively.

Let the acceleration of A be  $a_A$  and that of B be  $a_B$ . when the two bodies collide, the force on A is equal and opposite to the force on B, from Newton's third law; hence we can write:

$$F_A = -F_B \dots\dots\dots(10)$$

From Newton's second law,  $F = ma$  hence

$$m_A a_A = -m_B a_B \dots\dots\dots(11)$$

we recall that acceleration  $a$  is given by

$$a = \frac{v - u}{t}$$

$$a =$$

$$a =$$

Hence

$$m_A \frac{v_A - u_A}{t} = -m_B \frac{v_B - u_B}{t}$$

$$m_A (v_A - u_A) = -m_B (v_B - u_B) \dots\dots\dots(12)$$

$$m_A v_A - m_A u_A = -m_B v_B + m_B u_B$$

Where  $t$  is the contact time of the two bodies during collision and  $u$  is the initial velocity before collision and  $v$  is the final velocity after collision. Eliminating  $t$  from

$$m_A (v_A - u_A) = -m_B (v_B - u_B) \dots\dots\dots(13)$$

$$(m_A v_A + m_B v_B) = (m_A u_A + m_B u_B) \dots\dots\dots(14)$$

Since we have already defined momentum as mass  $\times$  velocity. The above equation means that the total momentum before the impact is equal to the total momentum after the impact

### Self-Assessment Exercise(s) 1

1. A 2.0 kg brick is moving at a speed of 6.0m/s. how large a force  $F$  is needed to stop the brick in a time of  $7.0 \times 10^{-4}$ s

2. An 8.0g bullet is fired horizontally into a 9.00kg cube of wood, which is at rest, and sticks in it. The cube is free to move and has a speed of 40cm/s after impact. Find the initial velocity of the bullet

#### 4.0 Conclusion

In this unit you should have learnt the definition mass and weight, similarities of mass and weight. In addition, force and motion and Newton's laws of motion

#### 5.0 Summary

Momentum (p) of a body is defined as the product of its mass and velocity

Impulse is the product of the force and time during which the force acts ( $I=ft$ ). impulse acts only for a short time.

Principle of the conservation of linear momentum states that when two or more momentum remains constant, provided there are no external forces acting on the bodies. Such a system of bodies is called an isolated system or closed system

In elastic collisions both the linear momentum and kinetic energy are conserved. In inelastic collisions, linear momentum is conserved, but kinetic energy decreases

#### 6.0 Tutor-Marked Assignments (TMAs)

1. Distinguish between impulse and momentum. How are they related? Calculate the momentum of:

- The moon of mass  $7 \times 10^{22}$  kg travelling with a velocity of 1.05km/s
- A tanker of mass  $7 \times 10^7$ kg travelling at 4m/s
- An aircraft of mass  $2.5 \times 10^5$ kg travelling at 500m/s

2. Distinguish between elastic and inelastic collision

3. State the law of conservation of linear momentum.

#### 7.0 References/Further Reading

#### Unit 5

Work, Energy and

Power

Content

#### 1.0 Introduction

#### 2.0 Learning Outcomes

#### 3.0 Learning Content

##### 3.1 concepts of work

##### 3.2 Concepts of energy

##### 3.3 Concepts of Power

#### 4.0 Conclusion

#### 5.0 Summary

#### 6.0 Tutor-Marked Assignment

#### 7.0 References/Further Reading

#### 1.0 Introduction

In this section we'll see how work is defined and how to calculate work in a variety of

situations involving constant forces. Even though we already know how to solve problems in which the forces are constant, the idea of work, energy and power is still useful in such problems.

## 2.0 Learning Outcomes

Upon completion of this unit, you should be able to:

- i. explain work, energy and power and give respective examples
- ii. calculate
  - a. The work done, given a force and displacement it produces in its direction
  - b. The gravitational potential energy at a height  $h$  above a given frame of reference
- iii. Calculate the power in watts given an applied force and the time to produce a displacement
- iv. Identify the type of energy possessed by a body under given conditions.
- v. State the law of conversation of energy

## 3.0 Learning Content

### 3.1 Grouping and Tabulation

#### 3.1.1 WORK

Work is said to be done whenever a force is applied to cause a displacement of a body.

When you push a car a certain distance, or you lift a load from the floor on to a table, or you stretch or compress an elastic body some distance, you are said to do work.

When no movement takes place after you have applied a force to a body, no work is done. You have done no work in carrying a bag for so long standing on a spot.

Work is said to be done whenever a force moves its point of application a distance in the direction of the force.

#### Calculation

In physics, work has a precise definition. When a force  $F$  (Newtons) acting on an object produces a small displacement  $S$  (metres), we say that the work done by the force on the object is given by the product of the force and displacement produced, namely,

$$W = F \times S \dots\dots\dots(1)$$

Unit:

$$\text{Joules} = \text{Newton} \times \text{metre} = \text{Joules}$$

Other units include :

$$\text{Kilojoules} = \text{KJ} = 10^3\text{J}$$

$$\text{Megajoules} = \text{MJ} = 10^6\text{J}$$

Consider the simple case of a straight-line motion, where a force  $\vec{F}$  acts on a body that simultaneously undergoes a vector displacement  $\vec{s}$ . The component of  $\vec{F}$  in the direction of  $\vec{s}$  is  $F\cos\theta$ . The work  $W$  done by the force  $\vec{F}$  is defined to be the component of  $\vec{F}$  in the direction of the displacement, multiplied by the displacement:

$$W = (F\cos\theta)(s) = Fs\cos\theta\dots\dots\dots(2)$$

Notice that  $\theta$  is the angle between the force and displacement vectors. Work is a scalar quantity.

If  $\vec{F}$  and  $\vec{d}$  are in the same direction  $\theta = 0^\circ$ ,  $\cos\theta = \cos 0^\circ = 1$  and  $W = Fs$ . But, if  $\vec{F}$  and  $\vec{d}$  are in opposite direction, then  $\cos\theta = \cos 180^\circ = -1$  and  $W = -Fs$ ; the work is negative. Forces such as friction often do negative work.

Example 5:

A UPS delivery man pushes on a box with a force of 100 N while sliding it along the floor a distance of 5 meters at constant speed. How much work is performed by the UPS man in moving the box?

Solution:

In this case the force and displacement are in the same direction so

$$W = F\cos\theta = (100\text{N})(5\text{m})\cos 0^\circ$$

$$\text{But } \cos 0^\circ = 1$$

$$W = 100 \times 5 = 500\text{Nm} = 500\text{J}$$

Work done in a force field

a. Lifting a body

Work is done against the force of gravity or against the weight of the body. We defined the weight of a body as the force of attraction on the body due to the earth's gravity.

The magnitude of such work done is given by

$$\text{Work} = \text{Force} \times \text{Distance}$$

$$= mg \times h \dots\dots\dots(3)$$

$$= mgh$$

Where  $m$  = mass of the body,  $g$  = acceleration due to gravity ( $g=10\text{m/s}^2$  and  $h$  = height

Example

A boy of mass 10kg runs up a set of steps in total height 3.0m. Find the work done against gravity

Solution

$$\text{Work done} = mgh$$

$$= 10 \times 10 \times 3$$

$$= 300\text{J}$$

b. Falling Bodies

When a body falls freely in a force field, the force of the field does work on the body.

Hence for a body falling freely under gravity the earth's gravitational force does work on the body. If the body of mass  $m$ , falls through a vertical height  $h$ , the work done by gravity on the body is given by  $w = mgh$

Example

A load sack of total mass 100kg falls down from the floor of a lorry 2.0m high. Calculate the work done by gravity on the load.

Solution

$$W = mgh$$

$$= 100 \times 10 \times 2$$

= 2000J

### 3.1.2 Energy

Energy is a very important concept both in physics and in our world at large. Energy takes various forms. A massive truck traveling along the highway at a high speed has much kinetic energy; a water reservoir just above a dam contains significant gravitational potential energy.

Energy is a measure of the change imparted to a system. It can be mechanically transferred to an object when a force does work on that body. The amount of energy given to an object via the action of the force over a distance equals the work done. Further, when an object does work, it gives up an amount of energy equal to the work it does. Because change can be effectuated in many different ways there are a variety of forms of energy. All forms of energy, including work, have the same units, joules. Energy is a scalar quantity. An object that is capable of doing work possess energy. There are many forms of energy. These include:

i. Mechanical Energy ii. Heat Energy iii. Light energy iv. Chemical energy v. Electrical energy vi. Atomic Energy vii. Solar energy

#### Types of Mechanical Energy

Mechanical energy is classified into two types – potential energy and kinetic energy

### 3.1.3 Kinetic Energy (K.E)

Kinetic Energy (K.E) is the energy possessed by an object because it is in motion. If an object of mass  $m$  is moving with a speed  $v$ , it has translational kinetic energy given by

1

$$K.E = \frac{1}{2}mv^2$$

#### Measurement of Kinetic Energy (K.E)

Kinetic energy depends only on the mass and velocity of the body. The kinetic (K.E) of a body in motion is given by:

1

$$K.E = \frac{1}{2}mv^2 \dots\dots\dots(4)$$

Thus if two bodies in motion have the same mass, the faster body has the greater kinetic energy and if they have the same velocity, the body with greater mass has a greater kinetic energy

#### Example

An object of mass 5kg is moving at a constant velocity 15m/s. Calculate its kinetic energy

#### Solution

1

$$K.E = \frac{1}{2}mv^2$$

1

$$K.E = \frac{1}{2} \times 5 \times 15^2 \text{ Joules}$$

2

$K.E = 562.5$  Joules

Where  $m$  is in kg and  $v$  is in m/s, the units of K.E are joules

Examples of Kinetic Energy are;

- i. A student running a race
- ii. An object falling freely under gravity
- iii. Wind or air in motion
- iv. Electrical charges in motion
- v. A moving bullet or a moving hammer head

### 3.1.4 Potential Energy

Potential Energy (P.E) is simply 'stored energy' or the energy possessed by a body by virtue of its position or state.

Such stored energy is used to do work when the body is free to move. A heavy stone on top of a table has potential energy. When allowed to fall on to a glass plate on the floor, it will shatter the plate. The potential energy of the stone due to its position above the floor is expended in shattering the plate. A body may have a potential energy due to its position in a force field. If the force field is the gravitational field, the body is said to possess a gravitational potential energy. The stone resting on top of a table has gravitational potential energy due to its height above the ground level. If the body is of mass  $m$  and the height of the table is  $h$ , then the gravitational potential energy is given by

$$E = mgh$$

Where  $m$  = mass of the body,  $g$  = acceleration due to gravity and  $h$  = height ( $g=10\text{m/s}^2$ )

### 3.1.5 Conservation of Energy

There is an important law of nature which governs these changes in forms of energy.

This law is known as the law of conservation of energy

The law of conservation of energy states that in an isolated or closed system, the total amount of energy is always constant, although energy may be changed from one form to another

By an isolated or closed system, we mean a group of objects that neither receives energy from or gives energy to objects outside the system. For example, the whole universe is an isolated system, because there is nothing outside it, the universe cannot share energy with things outside it

### 3.1.6 Power

The definition of work makes no reference to the passage of time. If you lift a barbell weighing 100 N through a vertical distance of 1.0 m at constant velocity, you do  $(100\text{N})(1.0\text{m}) = 100\text{J}$  of work whether it takes you 1 second, 1 hour, or 1 year to do it. But often we need to know how quickly work is done. We describe this in terms of power. In ordinary conversation the word "power" is often synonymous with "energy" or "force." In physics we use a much more precise definition: Power is the time rate at which work is done. Like work and energy, power is a scalar quantity.

Power is defined as the time rate of doing work

Work done by a force = .....

Work done = .....(5)

Power

If the work is in joules and the time is in seconds, then the power is measured in watts.

1 watt = 1 joule per second

Other commonly used units of power are the kilowatt (kW), the megawatt (MW) and the horsepower (h.p.),

1Kw = 1000W = 103 W

1MW = 1,000,000W = 106W

1 h.p. = 746W

Example

A car is moving at a constant speed of 20m/s. The force retarding its motion is 500N.

Calculate the engine power of the car required to maintain the motion.

Solution

Engine power = work done by the car per second in overcoming the retarding force

= force x distance moved per second

= 500N x 20m/s

= 10000Js-1

= 104 Watts or 10KW

Self-Assessment Exercise(s) 2

1. A man of mass 50kg ascends a flight of stairs 5m high in 5 seconds. If acceleration due to gravity is 10m/s<sup>2</sup>, the power expended is

A. 100W B. 200W C. 250W D. 400W E. 500W

2. What do you understand by the term work ? what is its unit?

3. A mass of 10kg moves with a velocity of 4m/s. find its kinetic energy

4.0 Conclusion

In this unit you should have learnt how to calculate the work done, given a force and displacement it produces in its direction., calculate the power in watts given an applied force and the time it takes to produce a displacement

5.0 Summary

1. Work is said to be done when a force F moves a body a distance, S in the direction of the force. Work is force x displacement in the direction of the force

2. Energy is the capacity to do work and is measured in joules.

3. The total energy in a closed or isolated system is constant. Thus in such a system, energy can never be created or destroyed but can be transformed from one form to another. This is the law of conservation of energy.

4. power is the time rate of doing work or the rate at which energy is expended. It is measured in Watts.

6.0 Tutor-Marked Assignments (TMAs)

A ball of mass 8kg falls from rest from a height of 100m. Neglecting air resistance,

calculate its total energy after falling a distance of 40m

A boy weighing 350N runs up a flight of stairs consisting of 20 steps each 10cm high in 10 seconds. What is his power ?

## 7.0 References/Further Reading

### Module 3

#### Circular Motion

##### Unit 1: Centripetal Force

##### Unit 2: Newtonian Gravitation

##### Unit 3: Circular Motion in Space

### Unit 1

#### Centripetal Force

##### Contents

##### 1.0 Introduction

##### 2.0 Learning Outcomes

##### 3.0 Learning Content

##### 3.1 Centripetal Force

##### 3.2 Banking of Tracks

##### 3.3 Motion along Horizontal and Vertical Circles

##### 4.0 Conclusion

##### 5.0 Summary

##### 6.0 Tutor-Marked Assignment (TMA)

##### 7.0 References/Further Reading

##### 1.0 Introduction

This unit introduces the concept of circular motion. There are several bodies or objects that exhibits circular motion such as movement of fan blades, movement of car wheels, movement of earth around the orbits etc.

##### 2.0 Learning Outcomes

At the end of this unit, you should be able to explain:

1. the inward force that keeps an object in circular motion
2. the speed and acceleration of an object in circular motion
3. the applications of circular motion

##### 3.0 Learning Content

##### 3.1 Centripetal Force

An important aspect of circular motion you should know is the type of force that keeps an object in circular motion. This force is called centripetal force. Centripetal force causes centripetal acceleration.

A body in circular motion moves with a constant speed but it has acceleration. The acceleration can be expressed mathematically as:

$a_c = \frac{v^2}{r}$

$a_c = \frac{4\pi^2 r}{T^2}$  .....(1)

where:

$a$  is the centripetal acceleration

$v$  is the linear speed

$r$  is the position of the object

From Newton's second law of motion:

$$F = ma \dots\dots\dots(2)$$

Thus, centripetal force is given as:

$$F = \frac{mv^2}{r}$$

$$F = \frac{mv^2}{r} \dots\dots\dots(3)$$

where:

$m$  is the mass of the body in circular motion

$r$  is the position of the object

An object in circular motion moves in an angular manner and has an angular displacement  $\Delta\theta$ , an angular speed  $\omega$  and an angular acceleration  $\alpha$ .

### Angular Displacement

Angular displacement  $\Delta\theta$  is the difference between the final angular displacement and the initial angular displacement of an object in circular motion. This is expressed mathematically as:

$$\Delta\theta = \theta_f - \theta_i \dots\dots\dots(4)$$

where:

$\theta_f$  is the final angular displacement

$\theta_i$  is the initial angular displacement

Angular displacement is expressed in radians or rads.

### Angular Speed

The angular speed is the rate of change of angular displacement of an object in circular motion. This is expressed mathematically as:

$$\omega = \frac{\Delta\theta}{\Delta t}$$

$$\omega = \frac{\Delta\theta}{\Delta t} \dots\dots\dots(5)$$

$$\Delta\theta$$

where:

$\omega$  is the angular speed

$\Delta\theta$  is the angular displacement.

$\Delta t$  is the change in time.

Angular speed is expressed in rad/s

### Angular acceleration

Angular acceleration is the rate of change in angular speed of an object in circular motion. This is expressed mathematically as:

$$\alpha = \frac{\Delta\omega}{\Delta t}$$

$$\alpha = \frac{\Delta\omega}{\Delta t} \dots\dots\dots(6)$$

where:

$\Delta\omega$  is the angular speed.

$\Delta t$  is the change in time.

Angular acceleration is expressed in  $\text{rad/s}^2$ .

The equation of linear motion with constant acceleration also applies to angular motion.

Table 3.1 Equations of motion for constant linear acceleration and constant angular acceleration

Example

A grindstone rotates at constant angular acceleration  $\alpha = 0.35 \text{ rad/s}^2$ . At time  $t = 0$ , it has an angular velocity of  $\omega_0 = -4.6 \text{ rad/s}$  and a reference line on it is horizontal at the angular position  $\theta_0 = 0$ . (a) At what time after  $t = 0$  is the reference line at the angular position  $\theta = 5.0 \text{ rev}$ ?

Solution

The angular acceleration is constant, so we can use the rotation equations of Table 3.1.

1

$$\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$$

2

Substituting known values and setting  $\theta_0 = 0$  and  $\theta = 5.0 \text{ rev} = 10\pi \text{ rad}$ , give us

$$10\pi = (-4.6)t + \frac{1}{2}(0.35)t^2$$

Solving this quadratic equation for  $t$ , we find

$$t = 32 \text{ s.}$$

Self-Assessment Exercise 1

1. A compact disc rotates from rest up to an angular speed of  $31.4 \text{ rad/s}$  in a time of  $0.892 \text{ s}$ . What is the angular acceleration of the disc, assuming the angular acceleration is uniform?

### 3.2 Banking of Tracks

When cars move along a curved path on roads and tracks, they incline at an angle. This angle is called a banking angle. Banking of tracks is associated with curved paths so that cars won't skid off the road. A centripetal force is required to keep cars moving along the path, that's why in banking of tracks, the outside edge of the track is made slightly higher than the inner edge.

Self-Assessment Exercise 2

1. A child's top is spun with angular acceleration  $\alpha = 5t^3 - 4t$  with  $t$  in seconds and  $\alpha$  in radians per second squared. At  $t = 0$ , the top has angular velocity  $5 \text{ rad/s}$ , and a reference line on it is at angular position  $\theta = 2 \text{ rad}$ . Obtain an expression for the angular velocity  $v(t)$  of the top

### 3.3 Motion along Horizontal and Vertical Circles

Consider a particle P (shown in the figure above) on a circle with radius  $r$  having an angle  $\theta$ . As P moves along the circle with time, the radius is constant. The particle has coordinates on both the x-axis and y-axis given by:

$$x = r \cos \theta, \quad y = r \sin \theta \quad \dots \dots \dots (7)$$

The radius  $r = OP$  and expressing this in vector form, we have:

$$\vec{r} = \vec{OP} = \vec{OP} \cos \theta + \vec{OP} \sin \theta \dots \dots \dots (8)$$

The velocity is given by:

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{v} = \frac{d}{dt} (\vec{OP} \cos \theta + \vec{OP} \sin \theta) \dots \dots \dots (9)$$

$$\vec{v} = \vec{OP} \left( \frac{d \cos \theta}{dt} + \frac{d \sin \theta}{dt} \right) \dots \dots \dots (10)$$

$$\vec{v} = \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right)$$

$$\vec{v} = \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right)$$

$$\vec{v} = \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right)$$

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$$\vec{v} = \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right)$$

$$\vec{v} = \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right)$$

$$\vec{v} = \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right)$$

Taking the magnitude of the velocity:

$$v = \left| \vec{v} \right|$$

$$v = \left| \vec{v} \right|$$

$$v = \left| \vec{v} \right|$$

$$v = \left| \vec{v} \right|$$

Acceleration is found by differentiating the expression for velocity. This gives:

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

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$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

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$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

$$\vec{a} = \frac{d}{dt} \left( \vec{OP} \left( -\sin \theta \frac{d\theta}{dt} + \cos \theta \frac{d\theta}{dt} \right) \right)$$

The magnitude of this expression is given as:

$$a = \left| \vec{a} \right|$$

$$a = \left| \vec{a} \right|$$

$\omega$

Thus:

$$\omega = \frac{v}{r}$$

Example

A car wheel has radius of 0.375m and moves at a constant speed of 16.6m/s. Calculate the angular speed of a stone attached to the wheel.

Solution:

$$\omega = \frac{v}{r}$$

$\omega$

$$= \frac{16.6}{0.375}$$

$\omega$

16.6

0.375

44.3  $\text{rad/s}$

$$\omega =$$

$\omega$

Self-Assessment Exercise 3

1. A car wheel has radius of 75cm and moves at a constant speed of 16.6m/s. Calculate the angular acceleration of a stone attached to the wheel.

#### 4.0 Conclusion

A body exhibiting circular motion requires a centripetal force to keep it moving. Consequently, this body will have an angular displacement  $\theta$ , angular speed  $\omega$ , and angular acceleration  $\alpha$ .

#### 5.0 Summary

In this unit, you have learnt the following:

The inward force that keeps an object moving in a circular motion. This is given by:

$$F_c = \frac{mv^2}{r}$$

$$F_c = m\omega^2 r$$

$\omega$

The parameters concerned with objects in a circular motion such as:

angular displacement:  $\Delta\theta = \theta_f - \theta_i$

$\Delta\theta$

angular speed  $\omega = \frac{\Delta\theta}{\Delta t}$

$\Delta t$

$\Delta\omega$

angular acceleration  $\alpha = \frac{\Delta\omega}{\Delta t}$

#### 6.0 Tutor-Marked Assignment (TMA)

1. A ball weighing 0.5 kg is tied to one end of a string and is whirled at a constant speed of  $10\text{ms}^{-1}$  in a horizontal plane. If the length of the string is 1 m, calculate the normal acceleration and the tension in the string.

2. A body of mass 1 kg tied to a string of length 1 m revolves in a horizontal circle. If the angular velocity is 3 rads<sup>-1</sup>, calculate the centripetal force acting on the body.

## 7.0 References/Further Reading

College Physics Cengage Learning Raymond A Serway. 2012.

Fundamentals of Physics Jearl Walker John Wiley & Sons Inc 2007

## Unit 2

### Newtonian Gravitation

#### Content

#### 1.0 Introduction

#### 2.0 Learning Outcomes

#### 3.0 Learning Content

#### 3.1 Gravitational Force, Inverse Square Law and Gravitational Constant

#### 3.2 Gravitational Potential and Field

#### 3.3 Galilean Invariance

#### 4.0 Conclusion

#### 5.0 Summary

#### 6.0 Tutor-Marked Assignment (TMA)

#### 7.0 References/Further Reading

#### 1.0 Introduction

In this unit, you will learn about the concept of Newtonian gravitation. The idea of gravitation was first discovered by Galileo. He performed several experiments by dropping objects from a high height and made inferences about how they drop to the ground. He concluded that neglecting air resistance, all objects fall at the same rate.

#### 2.0 Learning Outcomes

At the end of this unit, you should be able to:

1. describe how gravity acts on a free-falling object.
2. explain the concept of gravitational potential.
3. explain gravitational inverse square law.

#### 3.0 Learning Content

#### 3.1 Gravitational Force, Inverse Square Law and Gravitational Constant

Gravitational force is a type of force that causes attraction between two bodies. It is the force that attracts you towards the earth. The magnitude of gravitational force is dependent on two masses attracted to each other and the distance between them. Newton's inverse square law states that the force of attraction between two bodies is proportional to the product of the mass of the two bodies and inversely proportional to the square of the distance between them.

This law is mathematically expressed as:

$$F \propto \frac{1}{r^2}$$

$$F = \frac{G M m}{r^2}, \quad G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2} \text{ .....(1)}$$

$$r^2$$

where:

$G$  is the gravitational constant.

$F$  is the force of attraction

$m_1$  and  $m_2$  are the masses of the two bodies

$r$  is the distance between the two bodies.

### 3.2 Gravitational Potential and Field

A system of two bodies attracted to one another is associated with a gravitational potential energy. The gravitational potential energy associated with the system of the two objects of mass  $m_1$  and  $m_2$  at a distance  $r$  from the center of the Earth is expressed mathematically as:

$m_1$   $m_2$

$$U = -\frac{G m_1 m_2}{r} \quad (2)$$

where:

$U$  is the gravitational potential energy.

$m_1$  and  $m_2$  are the masses of the two bodies attracted to one another.

$r$  is the distance between the two bodies.

It is possible that a system in the gravitational field comprises more than two bodies.

For a three particle system, the gravitation potential energy of the system is given by:

$m_1$   $m_2$   $m_3$   $m_1$   $m_3$

$$U = -G \left( \frac{m_1 m_2}{r_{12}} + \frac{m_1 m_3}{r_{13}} + \frac{m_2 m_3}{r_{23}} \right) \quad (3)$$

$r_{12}$   $r_{23}$   $r_{13}$

An important thing you should note about bodies or objects not on Earth surface is that their gravitational potential energy decrease when the separation between them and the Earth surface decreases.

Example

Three 0.300 kg billiard balls are placed on a table at the corners of a right-angled triangle as shown in Figure 3.1. Find the net gravitational force on the cue ball (designated as  $m_1$ ) resulting from the forces exerted by the other two balls.

Solution:

Figure 3.1

To find the net gravitational force on the cue ball of mass  $m_1$ , we first calculate the force  $F_{21}$  exerted by  $m_2$  on  $m_1$ . This force is the y-component of the net force acting on  $m_1$ . Then we find the force  $F_{31}$  exerted by  $m_3$  on  $m_1$ , which is the x-component of the net force acting on  $m_1$ . With these two components, we can find the magnitude and direction of the net force on the cue ball.

First, find the magnitude of the force  $F_{21}$  exerted by  $m_2$  on  $m_1$  using the law of gravitation.

$m_2$   $m_1$

$$F_{21} = \frac{G m_2 m_1}{r_{21}^2}, \quad G = 6.67 \times 10^{-11}$$

$r_{21}$

$$\begin{aligned}
 &0.300 \times 0.300 \\
 &= 6.67 \times 10^{-11} \times \\
 &(0.400)^2 \\
 &F_{21} = 3.75 \times 10^{-11} \text{ N}
 \end{aligned}$$

Then, find the magnitude of the force  $F_{31}$  exerted by  $m_3$  on  $m_1$  using the law of gravitation,

$$\begin{aligned}
 &F_{31} = \frac{G m_3 m_1}{r^2} \\
 &= \frac{6.67 \times 10^{-11} \times 4.0 \times 6.0}{(0.300)^2} \\
 &F_{31} = 6.67 \times 10^{-11} \text{ N}
 \end{aligned}$$

The net force has components  $F_x = F_{31}$  and  $F_y = F_{21}$ .

Compute the magnitude of this net force:

$$\begin{aligned}
 F &= \sqrt{F_x^2 + F_y^2} = \sqrt{(3.75)^2 + (6.67)^2} \times 10^{-11} \\
 &= 7.65 \times 10^{-11} \text{ N}
 \end{aligned}$$

### Self-Assessment Exercise 1

1. Particle 1 with mass  $m_1 = 6.0 \text{ kg}$  and particles 2 and 3 of masses  $m_2$  and  $m_3 = 4.0 \text{ kg}$ , separated by a distance  $a = 2.0 \text{ cm}$ . What is the net gravitational force on particle 1 due to the other particles?

### 3.3 Galilean Invariance

Galilean invariance is a concept that is concerned with the theory of relativity. It deals with observing quantities such as position, time, velocity, acceleration, etc from a frame of reference.

The Galilean transformation for position is as follows:

$$x = x' + vt' \quad (4a)$$

$$t = t' \quad (4b)$$

$$y = y' \quad (4c)$$

The Galilean transformation for velocity is as follows:

$$\begin{aligned}
 v_x &= \frac{dx}{dt} \\
 &= \frac{dx' + v dt'}{dt} \quad (5a)
 \end{aligned}$$

$$\begin{aligned}
 v_x &= \frac{dx'}{dt} + v \\
 &= v_x' + v \quad (5b)
 \end{aligned}$$

$$v_y = v_y'$$

The Galilean transformation for acceleration is as follows:

$$\begin{aligned}
 a_x &= \frac{dv_x}{dt} \\
 &= \frac{dv_x'}{dt} \quad (6) \\
 a_x &= a_x'
 \end{aligned}$$

The inverse square law doesn't have a Galilean transformation.

#### 4.0 Conclusion

Two bodies can get attracted to one another and this is associated with the gravitational force between them.

#### 5.0 Summary

In this unit, you have learnt the following:

The force of attraction between two bodies. This is given by the inverse square law:

$$F = \frac{G m_1 m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

$$U = -\frac{G M m}{r}$$

The gravitational potential of two bodies attracted to one another is given by:

$$U = -\frac{G M m}{r}$$

$$U = -\frac{G M m}{r}$$

$$U = -\frac{G M m}{r}$$

#### 6.0 Tutor-Marked Assignment (TMA)

1. A solid sphere has a uniformly distributed mass of  $1 \times 10^4 \text{ kg}$  and a radius of  $1.0 \text{ m}$ . What is the magnitude of the gravitational force due to the sphere on a particle of mass  $m$  when the particle is located at a distance of  $1.5 \text{ m}$  from the center of the sphere?
2. What must the separation be between a  $5.2 \text{ kg}$  particle and a  $2.4 \text{ kg}$  particle for their gravitational attraction to have a magnitude of  $2.3 \times 10^{12} \text{ N}$ ?

#### 7.0 References/Further Reading

College Physics Cengage Learning Raymond A Serway. 2012.

Fundamentals of Physics Jearl Walker John Wiley & Sons Inc 2007

#### Unit 3

#### Circular Motion in Space

#### Content

#### 1.0 Introduction

#### 2.0 Learning Outcomes

#### 3.0 Learning Content

#### 3.1 Escape Velocity and Weightlessness

#### 3.2 Synchronous Orbit

#### 4.0 Conclusion

#### 5.0 Summary

#### 6.0 Tutor-Marked Assignment (TMA)

#### 7.0 References/Further Reading

#### 1.0 Introduction

In this unit, you will be introduced to circular motion in space. In space, satellites are examples of objects that exhibit circular motion because of their motion around planets. There are naturally occurring satellites like the moon and artificial satellites placed by Man in space.

## 2.0 Learning Outcomes

At the end of this unit, you should be able to:

1. calculate the escape velocity of the Earth.
2. classify and characterise orbiting satellites.

## 3.0 Learning Content

### 3.1 Escape Velocity and Weightlessness

It is possible to project an object upwards and it will not come down again. This happens when the object is projected to space with a speed called 'escape speed'. The escape speed of the Earth can be found from conservation of energy by the following expressions:

1 □ □

$$\frac{1}{2}mv^2 + \frac{GMm}{r} = \frac{1}{2}mv^2 - \frac{GMm}{\infty} \dots\dots\dots(1)$$

□

Since energy is conserved, the sum of the kinetic energy (K.E) and potential energy (P.E) equals zero

1 □ □

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = 0 \dots\dots\dots(2)$$

2 □

Making the speed v the subject of the formula:

2 □ □ □

$$v = \sqrt{\frac{2GM}{r}} \dots\dots\dots(3)$$

□ □

where:

G is the gravitational constant.

Me is the mass of the earth.

Re is the radius of the earth.

Example:

Find the escape speed for Earth if its mass is  $5.972 \times 10^{24}$  kg and its radius is 6371 km.

Solution:

Using the formula to find the escape speed for Earth, we have:

2 □ □ □

$$v = \sqrt{\frac{2GM}{r}}$$

□ □

$$= \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.972 \times 10^{24}}{6.37 \times 10^6}}$$

$$= 11.2 \text{ km/s}$$

### Self-Assessment Exercise 1

1. An asteroid, headed directly toward Earth, has a speed of 12 km/s relative to the planet when the asteroid is 10 Earth radii from the Earth's center. Neglecting the

effects of the Earth's atmosphere on the asteroid, find the asteroid's speed  $V_f$  when it reaches the Earth's surface.

### 3.2 Satellites

Satellites also exhibit circular motion. An orbiting satellite can be classified into size, location and shape. The size classification can be defined by the semi-major axis. The shape classification can be defined by elliptical and circular orbits.

For a satellite that orbits the Earth elliptically, both the speed and its distance from the Earth fluctuate with fixed periods.

For a satellite in an elliptical orbit with semi-major axis 'a', its mechanical energy can be expressed mathematically as:

□□□

$$\square = -2\square \dots\dots\dots(4)$$

where:

G is the gravitational constant.

M is the mass of the Earth.

m is the mass of satellite.

Example:

A playful astronaut releases a bowling ball of mass  $m = 7.20 \text{ kg}$  into a circular orbit about the Earth at an altitude  $h = 350 \text{ km}$ . What is the mechanical energy E of the ball in its orbit?

Solution

First, calculate the orbital radius

$$\begin{aligned} \square &= \square + h \\ &= 6370\text{km} + 350\text{km} = 6.72 \times 10^6 \text{ m} \end{aligned}$$

Now, you can calculate the mechanical energy

□□□

$$\square = -$$

$$2\square$$

$$6.67 \times 10^{-11} \times 5.972 \times 10^{24} \times 7.20$$

$$2 \times 6.72 \times 10^6$$

$$\square = -214 \text{ J}$$

### Self-Assessment Exercise 2

1. A playful astronaut releases a bowling ball of mass  $m = 7.20 \text{ kg}$  into circular orbit about the Earth at an altitude  $h = 350 \text{ km}$ . What is the mechanical energy  $E_0$  of the ball on the launchpad?

### 4.0 Conclusion

Conclusively, when an object is launched into space with an escape speed, it won't come down again. In this operation, energy is conserved.

### 5.0 Summary

In this unit, you have learnt the following:

■ The expression for escape speed of a body launched into space and this is given by:

$$v_{\text{esc}} = \sqrt{2gR}$$

$$v_{\text{esc}} = \sqrt{2gR}$$

$$v_{\text{esc}} = \sqrt{2gR}$$

■ The concept of an elliptical orbiting satellite defined by a semi-major axis. The energy is given by:

$$E = -\frac{GMm}{2a}$$

$$E = -\frac{GMm}{2a}$$

$$E = -\frac{GMm}{2a}$$

## 6.0 Tutor-Marked Assignment (TMA)

1. If the radius of Mercury earth is 2439 km and the acceleration due to gravity is 4ms<sup>-2</sup>. Calculate the escape velocity.

2. A 50 kg satellite circles planet Cruton every 6 hours. The magnitude of the gravitational force exerted on the satellite by Cruton is 80 N. What is the radius of the orbit?

## 7.0 References/Further Reading

College Physics Cengage Learning Raymond A Serway. 2012.

Fundamentals of Physics Jearl Walker John Wiley & Sons Inc 2007

## Module

### Rotation of Rigid Bodies

#### Unit 1: Rotational Variables

#### Unit 2: Moment of Inertia

#### Unit 3: Conservation of Angular Momentum

## Unit

### Rotational Variables

## Content

### 1.0 Introduction

### 2.0 Learning Outcome

### 3.0 Learning Content

#### 3.1 Angular Position

#### 3.2 Angular Velocity

#### 3.3 Angular Acceleration

### 4.0 Conclusion

### 5.0 Summary

### 6.0 Tutor – Marked Assignments

### 7.0 References/Further Reading

### 1.0 Introduction

In this unit, we discuss the rotation of rigid bodies. Rotation of rigid bodies occurs at all scales, from motions of atoms in elements to the motion of the entire planet. Rotating rigid bodies

cannot be adequately modelled as a moving point as each element of the body has its path as well as the speed of rotation. This module explains some general methods for analyzing the motion of a rotating rigid body.

## 2.0 Learning Outcomes

At the end of this unit you should be able to:

1. Determine the angular position, velocity and acceleration of rigid bodies.
2. Differential between the units of angular position, velocity and acceleration.

## 3.0 Learning Content

### 3.1 Angular Position

The position of a rigid body, rotating about a fixed axis, can be described by choosing a point P on the body and tracking the x and y coordinates of this point, as shown in figure 1. However, this method is not handy as it requires two variables to specify the position of the body.

Nevertheless, the line OP (from point P to a pivot point O as in Figure 3.1) is fixed in the body and rotates with it. The angle  $\theta$  that OP makes with the positive x-axis describes the angular position of the body.

Figure 3.1 A speedometer needle rotating about a fixed axis.

The angular position of a rigid body rotating around a fixed axis can be positive or negative.

Measuring the angles in a counterclockwise manner from the positive x-axis results in a positive angular position while measuring in a clockwise manner results in a negative angular position.

Furthermore, the most appropriate way to measure the rotation angle is not in degrees, but in radians. One radian (1 rad) is the angle subtended at the centre of a circle by an arc with a length equal to the radius of the circle. Given that s is the length on an arc and r is the radius as shown in figure 3.2, the value of  $\theta$  (in radians) is equal to:

$$\theta = \frac{s}{r}$$

$$\theta = 1$$

$$\theta = \frac{s}{r}$$

Figure 3.2 Measuring angles in radians.

### Self-Assessment Exercise 1

1. When is the angular position positive?

### 3.2 Angular Velocity

The time rate of change in the angular position of a rigid body is the angular velocity. The rotational motion of such a rigid can also be described in terms of the angular velocity. In figure 3.3, line OP makes an angle  $\theta_1$  with the positive x-axis at time  $t_1$  and makes an angle  $\theta_2$  with the positive x-axis at a later time  $t_2$ . The average angular velocity,  $\omega$ , in the time interval  $\Delta t = t_2 - t_1$  is expressed as:

$$\omega = \frac{\theta_2 - \theta_1}{\Delta t}$$

$$\omega = \frac{\Delta \theta}{\Delta t}$$

$$\omega = \frac{\Delta \theta}{\Delta t}$$

Note that OP in figure 3.3 is rotating about the z-axis, which is perpendicular to the plane of the diagram. The instantaneous angular velocity,  $\omega$ , is the limit of  $\frac{\Delta\theta}{\Delta t}$  as  $\Delta t$  approaches zero i.e. the derivative of  $\theta$  with respect to  $t$ :

$$\Delta\theta \rightarrow 0$$

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

$$\Delta\theta \rightarrow \Delta\theta \text{ (small)}$$

Figure 3.3 Angular displacement of a rotating body.

Also, note that anywhere angular velocity is mentioned, the instantaneous angular velocity is implied. In addition, the direction which the rigid body is rotating determines whether the angular velocity is positive or negative; The angular velocity is positive if the body is rotating in the direction of increasing angular position and negative if it is rotating in the direction of decreasing angular position. The unit of angular velocity is radian per second (rad/s). Other units such as revolutions per minute (rpm or rev/min) are also used.

Conversion between rad/s and rpm are as follows:

$$2\pi \text{ rad} = 1 \text{ rev}$$

$$1 \text{ rev} = \frac{2\pi \text{ rad}}{2\pi} = 1 \text{ rev}$$

$$60 \text{ s} = 1 \text{ min}$$

Angular velocity can be related to tangential speed through equation 1:

$$\omega = \frac{v}{r} \quad (5)$$

Taking the derivative of equation 5 with respect to time and noting that the radius  $r$  is a constant, results in:

$$\frac{d\omega}{dt} = \frac{dv}{dt} \cdot \frac{1}{r}$$

$$= \left(\frac{dv}{dt}\right) = a + \omega = \frac{v}{r} \quad (6)$$

$$\frac{d\omega}{dt} = \frac{dv}{dt} \cdot \frac{1}{r}$$

Where  $\frac{dv}{dt} = 0$  and is just the tangential speed  $v$ . Substituting equation 3 into 6 results in:

$$\omega = \frac{v}{r} \quad (7)$$

By extension, for constant angular velocity, the tangential speed of points in a rigid body increase with the increase in distance from the axis of rotation. Figure 3.4 shows two points on a circular rigid body, it can be implied that  $v_1 > v_2$  at constant  $\omega$  since  $r_1 > r_2$ .

Figure 3.4 Two points on a circular rigid body having different tangential speeds.

Self-Assessment Exercise(s) 2

1. For a rigid body, what happens to the tangential speed when the angular velocity is constant?
2. What is the unit of angular velocity?

### 3.3 Angular Acceleration

The time rate of change in the angular velocity of a rigid body is the angular acceleration i.e. the angular acceleration describes situations when the angular velocity changes (Figure 3.5). The average angular acceleration,  $\alpha$ , over an interval  $\Delta t = t_2 - t_1$  is expressed as:

$$\omega = \frac{\Delta\theta}{\Delta t}$$

$$\omega = 8$$

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Where  $\omega$  and  $\omega$  are the instantaneous angular velocities at times  $t$  and  $t$  respectively. The instantaneous angular acceleration,  $\alpha$ , is the limit of  $\frac{\Delta\omega}{\Delta t}$  as  $\Delta t$  approaches zero i.e. the derivative of  $\omega$  with respect to  $t$ :

$$\Delta\omega = \omega - \omega$$

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t}$$

$$\Delta \rightarrow \Delta\omega = \omega - \omega$$

Also, note that anywhere angular acceleration is mentioned, the instantaneous angular acceleration is implied. In addition, if the angular acceleration is positive, the angular velocity is increasing and if the angular acceleration is negative, then the angular velocity is decreasing. The unit of angular acceleration is radian per second per second ( $\text{rad/s}^2$ ).

### Example

A disc rotates such that it sweeps out an angle at the rate of  $\omega = 90^\circ \text{ per second}$ . The disc rotates counterclockwise when viewed in the plane of the page.

- What is the angular velocity of the disc?
- What direction is the angular velocity?
- How many radians does the disc rotate through in 45 s?
- What is the tangential speed of a point on the disc 5 cm from the axis of rotation?

Figure 3.5 Calculating the average angular acceleration of a rotating rigid body

### Solution

- Using equation 3,  $\omega$  can be differentiated with respect to time to get the angular velocity. Therefore,

$$\omega = \frac{\Delta\theta}{\Delta t}$$

$$\omega = (90^\circ) = 90^\circ (\text{per second})$$

$$\omega = 90^\circ \text{ per second}$$

$$\therefore \omega = 90^\circ \text{ per second}$$

- Since the disc is rotating in the direction of increasing angular position, the angular velocity is therefore positive.

- From equation 2,  $\Delta\omega = \omega - \omega$ . Then

$$\Delta\omega = \omega(45^\circ) - \omega(0^\circ) = 90(45^\circ) - 90(0^\circ)$$

$$\Delta\omega = 4050.0^\circ \text{ per second}$$

- From equation 7,  $\alpha = \frac{\Delta\omega}{\Delta t}$ . Then

$$\alpha = (0.1^\circ)(90^\circ \text{ per second})$$

$$\alpha = 9^\circ \text{ per second}$$

### Example 3.2

A bicycle quality inspector mounts a bicycle on a repair stand and spins the rear wheel from rest to a final angular velocity of 200 rpm in 10.00 s.

- Calculate the average angular acceleration in  $\text{rad/s}^2$ .

b. If he now hits the brakes, causing an angular deceleration of  $-83.76 \text{ rad/s}^2$ , how long does it take the wheel to stop?

Solution

a. Firstly, the change in angular velocity needs to be converted to  $\text{rad/s}$ . Using equation 4,

$$2\pi$$

$$\Delta\omega = 200 \frac{\text{rev}}{\text{min}} = 200 \frac{2\pi \text{ rad}}{60 \text{ s}}$$

$$60$$

$$\Delta\omega = 20.94 \frac{\text{rad}}{\text{s}}$$

Inserting this value into equation 8 gives:

$$\Delta\omega = 20.94 \frac{\text{rad}}{\text{s}}$$

$$\omega = 0$$

$$\Delta\omega = 10.0 \frac{\text{rad}}{\text{s}}$$

$$\alpha = -2.094 \frac{\text{rad}}{\text{s}^2}$$

b. From equation 8, it can be derived that  $\Delta\omega = \alpha \Delta t$ . Therefore,

$$0 - 20.94 \frac{\text{rad}}{\text{s}} = -2.094 \frac{\text{rad}}{\text{s}^2} \Delta t$$

$$\Delta t = 10.0 \text{ s}$$

$$\Delta\omega = -83.76 \frac{\text{rad}}{\text{s}^2} \Delta t$$

$$\Delta t = 0.25 \text{ s}$$

#### 4.0 Conclusion

Angular position, velocity and acceleration allow for the complete description of rotating rigid bodies.

#### 5.0 Summary

You have learnt in this unit the following:

- Angular position and its unit.
- Angular velocity and its unit.
- Angular acceleration and its unit.
- Using the formulas and equations to solve problems.

#### 6.0 Tutor-Marked Assignment

- A sprinter runs an 800-m race on a 400-m circular track in 60 s. What is his angular velocity assuming a constant speed?
- Calculate the angular velocity of Earth.
- A disc rotates at 750 rpm. If the diameter of the disc is 9 cm, (a) what is the tangential speed at a point at the edge of the disc?

#### 7.0 References/Further Reading

Hugh D. Young and Roger A. Freedman (2012) University Physics.

University Physics Volume 1 by OpenStax (2016).

Unit

Moment of Inertia

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## 1.0 Introduction

From unit 1 we discussed the rotational variables necessary to describe the rotation of rigid bodies. In this unit, we will discuss “Moment of Inertia”. It is the quantitative measure of rotational inertia, just as mass is the quantitative measure of linear inertia.

## 2.0 Learning Outcomes

At the end of this unit you should be able to:

1. Describe the differences between rotational and translational kinetic energy.
2. Describe the concept of moment of inertia.
3. Describe the parallel-axis theorem.

## 3.0 Learning Content

### 3.1 Moment of Inertia

The moment of inertia is the quantitative measure of rotational inertia, just as mass is the quantitative measure of linear inertia i.e. the more immense an object is, the more inertia it has, and the greater is its resistance to change in linear velocity. Similarly, the greater the moment of inertia of a rigid body or system of particles, the greater is its resistance to change in angular velocity about a fixed axis of rotation.

For a rigid body rotating about a fixed axis, the body can be thought of as being made up of a large number of particles, with masses at distances from the axis of rotation. The particles are labelled with the index  $i$ : The mass of the  $i$ th particle is  $m_i$  and its distance from the axis of rotation is  $r_i$ .  $r_i$  is the perpendicular distance from the rotation axis to the  $i$ th particle as the particles don't necessarily all lie in the same plane.

When a rigid body rotates about a fixed axis, the speed of the  $i$ th particle is given by equation 7,  $v_i = r_i \omega$ , where  $\omega$  is the body's angular speed. Different particles, within the rotating body, have different values of  $r_i$ , but  $\omega$  is the same for all. The kinetic energy of the  $i$ th particle can be expressed as

1 1

$$K_i = \frac{1}{2} m_i v_i^2 = \frac{1}{2} m_i r_i^2 \omega^2$$

2 2

The total kinetic energy of the body is the sum of the kinetic energies of all the particles:

1 1 1

$$K = \frac{1}{2} m_1 r_1^2 \omega^2 + \frac{1}{2} m_2 r_2^2 \omega^2 + \dots = \frac{1}{2} I \omega^2$$

2 2 2

Taking the common factor  $\frac{1}{2}$  out of this expression:

1 1

$$I = (\frac{1}{2} m_1 r_1^2 + \frac{1}{2} m_2 r_2^2 + \dots) = \frac{1}{2} \sum m_i r_i^2$$

2 2

The quantity in parentheses, obtained by multiplying the mass of each particle by the square of its distance from the axis of rotation and adding these products, is denoted by  $I$  and is called the moment of inertia of the body for this rotation axis:

$$I = \frac{1}{2} m_1 r_1^2 + \frac{1}{2} m_2 r_2^2 + \dots = \frac{1}{2} \sum m_i r_i^2$$

The SI unit of moment of inertia is the  $\text{kg} \cdot \text{m}^2$ . The rotational kinetic energy  $K$  of a rigid body is described in terms of the moment of inertia  $I$  as

1

$$K = \frac{1}{2} I \omega^2$$

2

The rotational and translational quantities for kinetic energy and inertia are summarized in table 1.

Table 3.1 Rotational and translational kinetic energies and inertia

Rotational Translational

$$K = \frac{1}{2} I \omega^2$$

1 1

$$K = \frac{1}{2} I \omega^2 = \frac{1}{2} M v^2$$

2 2

Furthermore, figure 3.1 gives values of rotational inertia for common object shapes around specified axes.

Self-Assessment Exercise(s) 1

1. What is the justification for the moment of inertia?
2. What is the SI unit of moment of inertia?

Figure 3.1 Rotational inertia for common object shapes.

Example 3.1

A machine part (figure 3.2) consists of three disks linked by light-weight struts.

- a. What is this body's moment of inertia about an axis through the centre of disk A, perpendicular to the plane of the diagram?
- b. What is its moment of inertia about an axis through the centres of disks B and C?
- c. What is the body's kinetic energy if it rotates about the axis through A with angular speed  $\omega = 3.0 \text{ rad/s}$ ?

Figure 3.2 An unusually shaped machine part

Solution

- a. The particle at point A lies on the axis through A, so its distance  $r$  from the axis is

zero and it, therefore, contributes nothing to the moment of inertia. Hence only B and C contribute, and equation 3.4 gives

$$I = I_{\text{cm}} = (0.20 \text{ kg})(0.50 \text{ m})^2 + (0.40 \text{ kg})(0.40 \text{ m})^2$$

$$I = 0.114 \text{ kg} \cdot \text{m}^2$$

b. The particles at B and C both lie on axis BC, so neither particle contributes to the moment of inertia. Hence only A contributes:

$$I = I_{\text{cm}} = (0.50 \text{ kg})(0.40 \text{ m})^2$$

$$I = 0.08 \text{ kg} \cdot \text{m}^2$$

From 3.5,

$$I = I_{\text{cm}} + I_{\text{cm}}$$

$$I = I_{\text{cm}} = (0.114 \text{ kg} \cdot \text{m}^2 + 0.08 \text{ kg} \cdot \text{m}^2)(3.0 \text{ m/s}^2)$$

$$I = 0.342 \text{ kg} \cdot \text{m}^2$$

### 3.2 Parallel-Axis Theorem

A body doesn't have just one moment of inertia, but it has infinitely many because there are infinitely many axes about which the body can rotate. There is a relationship between the moment of inertia of a body of mass  $M$  about an axis through its centre of mass and the moment of inertia about any other axis parallel to the original one but displaced from it by a distance  $d$ . This relationship is called the parallel axis theorem. It states that

$$I = I_{\text{cm}} + Md^2 \quad 6$$

Where  $I_{\text{cm}}$  is the moment of inertia about the axis through the centre of mass and  $I$  is the moment of inertia about the axis through point P, parallel to the axis through the centre of mass.

### Self-Assessment Exercise 2

1. What is the justification for the parallel-axis theorem?

#### Example 3.2

Calculate the moment of inertia of the thin rod in figure 3.1e, where the  $y$  axis is located at one end of the rod.

**Solution**

From figure 3.1e, for a thin rod,  $I_{\text{cm}} = \frac{1}{12}ML^2$ .

Therefore, at one end of the rod:

$$I = I_{\text{cm}} + Md^2$$

$$I = \frac{1}{12}ML^2 + ML^2 = \frac{13}{12}ML^2$$

$$I = \frac{13}{12}ML^2$$

$$I = \frac{13}{12}ML^2$$

$$I = \frac{13}{12}ML^2$$

$$I = \frac{13}{12}ML^2$$

$$I = \frac{13}{12}ML^2$$

$$I = \frac{13}{12}ML^2$$

$$I = \frac{13}{12}ML^2$$

#### 4.0 Conclusion

For a system of point particles rotating about a fixed axis, the moment of inertia for the system is  $I = \sum m_i r_i^2$ , where  $m_i$  is the mass of the point particle and  $r_i$  is the distance of the point particle to the rotation axis. The moment of inertia is the rotational counterpart to the mass in linear motion. The parallel-axis theorem is as a result of the possibility of a body to have multiple moments of inertia, as well as multiple parallel axes of rotation.

#### 5.0 Summary

You have learnt in this unit the following:

1. The moment of inertia for a system of point particles rotating about a fixed axis is:

$$I = \sum m_i r_i^2$$

2. The parallel axis theorem states that:

$$I = I_{cm} + M d^2$$

#### 6.0 Tutor-Marked Assignment

- i. Calculate the angular velocity of the orbital motion of Earth around the Sun.
- ii. A motorist is riding such that the wheels of the bike have a rotation rate of  $100 \text{ rev/min}$ . If the motorist brakes such that the rotation rate of the wheels decreases at a rate of  $1.4 \text{ rev/s}$ , how long does it take for the motorist to come to a complete stop?

#### 7.0 References/Further Reading

Hugh D. Young and Roger A. Freedman (2012) University Physics.  
University Physics Volume 1 by OpenStax (2016).

Unit

Angular Momentum

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1.0 Introduction

Angular momentum is the rotational equivalent of linear momentum. Any rigid body that rotates about an axis carries angular momentum, including rotating discs, planets, stars, etc. The concept of conservation of angular momentum as well as rolling bodies is discussed later in this unit.

## 2.0 Learning Outcomes

At the end of this unit you should be able to:

1. Describe the vector nature of angular momentum.
2. Calculate the angular momentum of a rigid body rotating about a fixed axis.
3. Describe the physics of rolling motion without slipping.

## 3.0 Learning Content

### 3.1 Angular Momentum of a Single Particle

The angular momentum  $\vec{L}$  of a particle is defined as the cross-product of the particle's position  $\vec{r}$  and linear momentum  $\vec{p}$ . It is also perpendicular to the plane containing  $\vec{r}$  and  $\vec{p}$ :

$$\vec{L} = \vec{r} \times \vec{p} \quad 1$$

The magnitude of the single-particle angular momentum is:

$$L = rp \sin \theta \quad 2$$

Where  $\theta$  is the angle between  $\vec{r}$  and  $\vec{p}$ . The unit of angular momentum is  $\text{kg} \cdot \text{m}^2/\text{s}$ .

Furthermore, taking the time derivative of the angular momentum gives an expression for the torque on the particle:

$$\begin{aligned} \frac{d\vec{L}}{dt} &= \frac{d}{dt} (\vec{r} \times \vec{p}) \\ &= \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} \\ &= \vec{v} \times \vec{p} + \vec{r} \times \vec{F} \\ &= \vec{0} + \vec{r} \times \vec{F} \\ &= \vec{r} \times \vec{F} \end{aligned}$$

From Newton's second law,

$$\begin{aligned} \frac{d\vec{L}}{dt} &= \vec{r} \times \vec{F} \\ &= \vec{r} \times m\vec{a} \end{aligned}$$

And it extends from the definition of torque that:

$$\begin{aligned} \frac{d\vec{L}}{dt} &= \vec{\tau} \end{aligned}$$

### Self-Assessment Exercise(s) 1

1. What is the magnitude of the angular momentum of a single particle?
2. What is the SI unit of angular momentum?

### Example 3.1

Determine the total angular momentum due to the three particles (in figure 3.1) about the origin.

**Solution**

The position vector and linear momentum vector need to be calculated first to be able to

calculate the angular momentum:

Particle 1;

$$\vec{r} = -2.0\hat{i} + 2.0\hat{j}$$

$$\vec{p} = (3.0\hat{i})(2.0\hat{j}/\text{s}) = 6.0\hat{i} \cdot \text{kg}\cdot\text{m}/\text{s}$$

$$\vec{L} = (-2.0\hat{i} + 2.0\hat{j}) \times (6.0\hat{i} \cdot \text{kg}\cdot\text{m}/\text{s}) = -12.0\hat{k} \cdot \text{kg}\cdot\text{m}^2/\text{s}$$

Particle 2;

$$\vec{r} = 3.0\hat{i} + 1.0\hat{j}$$

$$\vec{p} = (5.0\hat{i})(2.0\hat{j}/\text{s}) = 10.0\hat{i} \cdot \text{kg}\cdot\text{m}/\text{s}$$

$$\vec{L} = (3.0\hat{i} + 1.0\hat{j}) \times (10.0\hat{i} \cdot \text{kg}\cdot\text{m}/\text{s}) = -10.0\hat{k} \cdot \text{kg}\cdot\text{m}^2/\text{s}$$

Figure 3.1 Three particles in the xy-plane with different position and momentum vectors.

Particle 3;

$$\vec{r} = 2.0\hat{i} - 1.0\hat{j}$$

$$\vec{p} = (7.0\hat{i})(1.5\hat{j}/\text{s}) = 10.5\hat{i} \cdot \text{kg}\cdot\text{m}/\text{s}$$

$$\vec{L} = (2.0\hat{i} - 1.0\hat{j}) \times (10.5\hat{i} \cdot \text{kg}\cdot\text{m}/\text{s}) = 10.5\hat{k} \cdot \text{kg}\cdot\text{m}^2/\text{s}$$

Therefore,

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \vec{L}_3 = -11.5\hat{k} \cdot \text{kg}\cdot\text{m}^2/\text{s}$$

### 3.2 Angular Momentum of a Rigid Body

For a rigid body rotating around an axis of symmetry, its angular momentum  $\vec{L}$  and angular velocity  $\vec{\omega}$  are in the same direction. The vector relationship between them is:

$$\vec{L} = I\vec{\omega}$$

Also, if the total angular momentum of the system of particles is  $\vec{L}$  and the sum of the external torques is  $\sum \vec{\tau}$  then,

$$\frac{d\vec{L}}{dt} = \sum \vec{\tau}$$

$$\vec{L} = 7$$

$$\vec{L} = 7$$

#### Example 3.1

A radiator fan in a sports car has a moment of inertia of  $8\text{ kg}\cdot\text{m}^2$  about its axis of rotation. The angular velocity of the fan is  $\omega = (25\text{ rad/s})\hat{k}$  at the start. Find the fan's angular momentum as a function of time find its value at  $t = 2.0\text{ s}$

Solution

$$\vec{L} = I\vec{\omega} = (8\text{ kg}\cdot\text{m}^2)(25\text{ rad/s})\hat{k}$$

$$\vec{L} = (200\text{ kg}\cdot\text{m}^2/\text{s})\hat{k}$$

At  $t=2.0\text{ s}$ ,

$$\vec{L} = 800\text{ kg}\cdot\text{m}^2/\text{s}$$

### 3.3 Conservation of Angular Momentum

The law of conservation of angular momentum states that:

when the net external torque acting on a system is zero, the total angular momentum of the system is constant (conserved).

Mathematically, if

$$\frac{dL}{dt} = 0$$

Then,

$$\frac{dL}{dt}$$

$$= 0$$

$$\frac{dL}{dt}$$

And  $L$  is a constant. In other words, the total angular momentum of the system is constant.

#### 4.0 Conclusion

In further units, we shall study the tangent and normal to conic sections, tangent and normal are straight lines, therefore, studying the equation of straight line becomes necessary for further studies.

#### 5.0 Summary

You have learnt in this unit the following:

- Any rigid body that rotates about an axis carries angular momentum.
- The angular momentum  $\vec{L}$  of a particle is defined as the cross-product of the particle's position  $\vec{r}$  and linear momentum  $\vec{p}$ .
- Taking the time derivative of the angular momentum gives an expression for the torque on the particle.
- The angular momentum  $L$  of a rigid body rotating around an axis of symmetry is given by  $L = I\omega$ .
- The law of conservation of angular momentum states that: when the net external torque acting on a system is zero, the total angular momentum of the system is constant (conserved).

#### 6.0 Tutor-Marked Assignment

- What is the angular velocity of a 1m-diameter tire on an aeroplane travelling at 220.0 km/h?
- A man rides his scooter 500 m. The wheels have radius 15.0 cm. What is the total angle the tires rotate through during his trip?
- A rigid body with a cylindrical cross-section is released from the top of a 45° incline. It rolls 7.5 m to the bottom in 5.0 s. Find the moment of inertia of the body in terms of its mass  $m$  and radius  $r$ .

#### 7.0 References/Further Reading

Hugh D. Young and Roger A. Freedman (2012) University Physics.  
University Physics Volume 1 by OpenStax (2016).